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CYBERSECURITY RELATED STANDARDS MANUAL

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The North American Energy Standards Board (NAESB), Wholesale Gas Quadrant (WGQ) Version 4.0 Standards Manuals Relating to Additional Standards, Nominations, Flowing Gas, Invoicing, Quadrant Electronic Delivery Mechanisms, Capacity Release, Contracts, Interpretations, and WGQ (REQ / RGQ Version 4.0) Internet Electronic Transport, WGQ Electronic Tariff Filing (WEQ Version 4.0), Cybersecurity and any amendments or errata thereto, are protected by NAESB’s federal copyright 1996-2023. NAESB hereby grants the authorized users who are NAESB members in good standing permission to reproduce material therein for internal reference and use and not for use by any unauthorized third parties. Reproduction in any other form, or for any other purpose, is forbidden without express permission of NAESB. Copies are available for purchase from NAESB. This non-exclusive limited license is non-transferable and may be revoked without notice upon violation of the terms contained herein or any applicable law or regulation. Each user grants NAESB the right to audit its use to assure compliance with these terms.

TABLE OF CONTENTS

Section	Page
EXECUTIVE SUMMARY	4
KEY ASSUMPTIONS.....	6
INTRODUCTION.....	7
BUSINESS PROCESS AND PRACTICES	9
RELATED STANDARDS	20
TECHNICAL IMPLEMENTATION.....	21
TESTING GUIDELINES.....	47
APPENDICES	49
VERSION NOTES.....	57



EXECUTIVE SUMMARY

This North American Energy Standards Board (NAESB) Wholesale Gas Quadrant (WGQ) has developed standards for secure, electronic commerce over the Internet. The Cybersecurity Related standards include the use of Internet Electronic Transport (Internet ET), which enables the rapid, reliable and safe transportation of electronic information between NAESB Trading Partners.

This document is a high-level guide to implementing various technologies necessary to communicate transactions and other electronic data using standard protocols. As such, this guide is not intended to be a comprehensive, in-depth manual. Where possible, this guide points to more in-depth material. The Reference section provides locations on the Internet to obtain more information as well as recommended books and periodicals.

BUSINESS REASONS FOR USING INTERNET ET

Energy companies need to exchange information and data with other energy companies. Internet ET enables this with the following advantages:

Security. Internet ET incorporates the PAIN security principles of Privacy, Authenticity, Integrity and Non-repudiation.

Standardized Process. Internet ET standardizes how packages are exchanged, regardless of the business process, or the Trading Partner.

Audit Trail. Internet ET gives both Sender and Receiver a detailed audit trail, enabling better controls and less errors.

Error Notification. Internet ET prescribes how errors are to be handled, and provides a foundation for efficient and quick resolution to errors.

Minimum technology requirements. Internet ET is built on low-cost technology and readily-available Web browser and open source technology.

Interactive and Batch Capabilities. Internet ET provides mechanisms for both fully-automated and manual-assisted business processes.

Any Payloads. Internet ET can deliver any kind of payload, whether it is EDI, flat-files, XML, documents, etc.

Software Standards. The Internet ET standards increase the likelihood that software vendors will provide Commercial Off-The-Shelf (COTS) software packages.

OVERVIEW OF ELECTRONIC TRANSPORT LIFE CYCLE

In the Internet ET life-cycle, the party sending data, the 'Sender', creates an Electronic Package by encrypting the data payload and applying appropriate header 'envelope' information such as 'to' and 'from'. This Electronic Package is submitted to the Trading Partner's SSL Web server as an HTTP Request using the POST method.

The Receiving Party, the 'Receiver', receives and decrypts the package, then forwards the payload data to back-office processes. A Receipt is sent from the Receiver to the Sender with timestamps and any error notices. The Receiver's back-office systems process the data according to NAESB WGQ Quadrant Electronic Delivery Mechanisms (QEDM)-specific standards (e.g. 'Nominations'), Trading Partner Agreements, and related documents. If the Receiver decrypts in a separate process, the Receiver may send an Error Notification package to the Sender to identify errors found during decryption.

Trading Partners can be either the Sender or Receiver depending on what information and data needs to be exchanged.

The Internet ET standards focus on the transport of the Electronic Package and not the contents of the package. Each business process may define different contents (e.g. EDI, flat files, etc), and the Internet ET is designed to work with any type.

The following are Internet ET life-cycle scenarios:

1. **Success.** The Successful scenario is when the Electronic Package was delivered with no errors, and the Sender has received a Receipt from the Receiver.
2. **Invalid Package Response.** The Invalid Package Response scenario is when the Receiver was unable to disassemble the Electronic Package, and has sent an HTTP Response to the Sender notifying them of package errors.
3. **Invalid Package Error Notification.** The Invalid Package Error Notification scenario is when a Receiver detects an error in the package AFTER the Response is sent. This scenario exists when a Receiver has implemented processes where the decryption occurs after the Response is sent. Decryption errors are communicated to the Sender via an HTTP Request using the Internet ET Error Notification format.
4. **Exchange Failure.** The Exchange Failure scenario is when a Sender is unable to establish and/or maintain a connection with the Server to send an Electronic Package to the Receiver.

Errors detected after successful decryption (e.g. format errors, EDI errors, etc) are beyond the scope of the Internet ET, and can be found in the WGQ QEDM standards.

Parties implementing Internet ET should become familiar with the following components of the Internet ET:

- Internet ET Network and Communications Requirements
- Sending Internet ET Electronic Packages
- Receiving Internet ET Electronic Packages
- Security

KEY ASSUMPTIONS

This document makes the following assumptions:

- **Platform Independence.** An Internet ET implementation can communicate with all Trading Partners in the energy industry, regardless of what hardware, operating system and programming languages Trading Partners use.
- **Open Standards.** NAESB has adopted open standard technologies to provide flexibility and scalability.
- **Payload Content Independence.** Internet ET standards focus on the transport of the Electronic Package, and not the contents of the package. Each business process may define different contents. Internet ET is designed to work with any type of content (e.g. EDI, flat files, etc). The Internet ET's main function is to get the package from point X to point Y reliably with privacy, authentication, integrity, and non-repudiation.
- **Importance of the Trading Partner Worksheet (TPW).** Internet ET relies on the exchange of technical information between Trading Partners to establish and maintain reliable Internet ET production. This worksheet is intended to establish communications between two parties. Additional requirements and information may be required per the WGQ QEDM manual. The TPW is included with the NAESB Trading Partner Agreement (TPA). The TPW and TPA constitute the entire understanding between the Trading Partners regarding EDI.
- **Testing With Internet ET Trading Partners.** Since the Internet ET is not platform-specific, testing with other Trading Partners on a variety of platforms is very important in ensuring that each Internet ET application is compatible with a range of platforms used by various Trading Partners. Testing should ensure receipt of the package, proper decryption, and appropriate Receipts were sent.
- **Business Process Considerations.** Implementers of business processes that use Internet ET should be aware of the following issues that may impact business process design:
 1. **The Internet Lacks Quality of Service (QoS).** The Internet is unable to assign priority to file transfers. High-priority NAESB Internet ET package transfers such as Nominations have no priority over low-priority Internet transfers such as music MP3 files or other lower-priority NAESB Internet ET transfers. Business processes that have firm or tight Internet ET transfer timing requirements should be constructed to properly mitigate the risk associated with this lack of guaranteed QoS on the Internet. QoS may be improved by using a private network in lieu of the Internet.
 2. **Clock Synchronization.** The Internet ET allows +/- 5 seconds variance from a NIST atomic clock. Business processes with more stringent requirements may need to implement more restrictive synchronization requirements and processes.
 3. **Exchange Failures.** When Trading Partner's systems are failing, parties are required to attempt to send Internet ET packages 3 times over a minimum period of 30-minutes before notifying Trading Partners of Exchange Failures. Business processes with more stringent requirements may need to implement more restrictive Exchange Failure requirements and processes.

Examples Provided in this Document. The examples provided in this document are for illustration only. Implementers should rely on the standards and not on these examples when implementing the Internet ET.

INTRODUCTION

The North American Energy Standards Board (NAESB) is a voluntary non-profit organization comprised of members from all aspects of the greater gas and electric industries.

The Cybersecurity Related Standards manual, formerly known as Internet Electronic Transport (Internet ET) Standards manual, is used by the Wholesale Gas Quadrant (WGQ) for the secure, electronic transport of transactions and other information payloads between Trading Partners.

Recognizing timely cybersecurity standards updates are typically necessary to mitigate cybersecurity threats, the WGQ determined that consolidating cybersecurity related standards and text to a single WGQ manual would assist the rapid adoption through both the NAESB standards-setting and FERC regulatory processes. Since the majority of security specific standards already existed in the Internet ET manual, in 2022, the WGQ determined to copy all Internet ET related standards into a new Cybersecurity Related Standards manual, replacing the Internet ET manual. Additionally, the WGQ reviewed all WGQ implementation guides, and relocated security-specific standards to the new Cybersecurity Related Standards manual.

Version 4.0 of the Cybersecurity Related Standards manual incorporates all electronic transport technical specifications of the NAESB WGQ Internet ET Version 3.2. For full text of these modifications, refer to the Final Actions for NAESB WGQ Version 4.0 on NAESB's home page.

As the energy industry evolves and uses NAESB standards, additional and amended NAESB standards will be necessary. Any industry participant seeking additional or amended standards (including principles, definitions, standards, data elements, process descriptions, technical implementation instructions) should submit a request detailing the change to the NAESB office so that the appropriate process may take place to amend the standards.

Executive Summary

Provides a brief outline of the industry business situation which is the basis for development of this guide.

Introduction

Provides a background statement about NAESB's Mission and the underlying concepts behind the design and use of this guide.

Business Process & Practices

Provides a brief overview of the business process and the NAESB-approved principles, definitions and standards related to the business process covered by this guide.

Related Standards

Provides a reference to any related standards.

Technical Implementation – Internet Electronic Transport (Internet ET)

Provides an overview of the business process for Internet ET.

Data Dictionary

Provides definition of the standard data elements and the usage requirements for each element.

Batch Flow Diagram

Sending Electronic Packages

Provides instructions to develop mechanisms for sending of NAESB standard format data files.

Receiving Electronic Packages

Provides instructions to develop mechanisms for receiving of NAESB standard format data files.

Security

Provides guidelines for data privacy, data integrity, authentication and non-repudiation of inbound and outbound packages.

Other Considerations

Provides information regarding Error Notification and testing. Includes a reference guide and examples for repudiation and validation.

Testing Guidelines

Provides guidelines for testing the Internet ET standards.

Appendices

Table 1 – Internet ET Standard Error Codes and Messages

Appendix A – Reference Guide

Appendix B – Minimum Technical Characteristics and Guidelines for the Developer and User of the Customer Activities and Informational Postings Web Sites

Appendix C – Minimum Technical Characteristics for EDM Communications

Appendix D – Frequently Asked Questions

Version Notes

Contains a brief summary of modifications for each version.

BUSINESS PROCESS AND PRACTICES

A. OVERVIEW

Role of Internet Electronic Transport (Internet ET) in NAESB WGQ Business Processes

Business processes defined by NAESB Quadrants require the exchange of transactions and transaction data. The Internet ET, in concert with the WGQ QEDM, enables NAESB parties to securely and reliably exchange transactions over the Internet. Internet ET electronic 'packages' are created using the standards defined in this document.

Roles in Internet ET

In the Internet ET life-cycle, one party sends a package, and the other party receives the package. The party sending the package is referred to as the Sender or Client, and the party receiving the package is also referred to as the Receiver or Server.

NAESB business processes often require that parties act in both the Sender and Receiver roles. For example, once the Receiver of a payload file of Nominations has successfully processed the payload, they switch to the Sender role to send Nomination acknowledgements back to the original Sender. Internet ET implementations may need to implement both Sender and Receiver capabilities.

The standards adopted for Internet ET should be adhered to by the trading parties as minimum standards. A trading party may offer additional functions or features as options but should not require their use. Such additional features or functions are termed 'mutually agreed to'. If both Trading Partners agree on the inclusion, the additional feature requirements will be met. If either trading party does not agree to the inclusion of additional features, then the partners must allow for transmission and receipt of data using the minimum standards.

To establish an Internet ET trading partnership with another company, a company needs to exchange technical information about their Internet ET implementation. This may include:

- Contact information
- Public Keys, including key exchange and update policies
- Test Uniform Resource Locator (URL)s
- Production URLs, including alternative paths if available
- Common Code Identifiers (e.g. DUNS number)

This may be exchanged using the TPW.

Implementation Approaches

The NAESB Internet ET can be constructed using any IT deployment model, including the use of in-house development, consulting/development help from a Third-Party, Commercial Off-The-Shelf (COTS) software, or an outsourced solution with a Third-Party. The best solution for each organization must be determined based on the assessment of specific needs and the resources available to that organization.

All parties should fully investigate the ramifications of implementing electronic commerce using the Internet. This includes ensuring that all customer data, internal data, and applications are secured from intruders or other unauthorized parties.

Participation in electronic commerce over the Internet involves hardware, software, and technical expertise. Hardware requirements may include a server to receive incoming Internet ET packages and a firewall to block intruder access. Software includes operating software for the servers, including the firewall, programming languages which support Internet technologies, and encryption/decryption software to provide security during the transfer. Technical expertise may be involved in the development and maintenance of server applications to process incoming files as well as applications to initiate communication with a Trading Partner's server.

Internet ET Network and Communication Requirements

Trading Partners should maintain redundant connections to the public Internet for Internet ET sites. These redundant connections should be topographically diverse paths to minimize the probability of a single point of failure. Three possible approaches to redundant connections are:

1. Maintain multiple ISPs and multiple points of connectivity, each of which was identified by the same URL making the process of redundancy transparent to the Sender.
2. Maintain different Internet connectivity URLs (presumably on topographically different ISPs). For this to result in communication redundancy, the Sender should know of the existence of the secondary URL and have programming in place that will automatically switch Batch Browser transmissions to the secondary URL when the primary URL is unavailable.
3. Maintain multiple connections to the same ISP. This involves only one URL but the presumption would be that the ISP would provide alternate diverse paths for the URL.

Servers may maintain multiple URLs and, if such have been disclosed, the Sender should attempt to use these during primary URL outages. The redundant public Internet connections can be through a single ISP or multiple ISPs. If multiple URLs are provided for Internet ET access, the following conditions should be met:

- The information provided by each URL should be exactly the same, although the 'trans-id' sequences may differ.
- The Trading Partners should be informed of both URLs and their availability.
- The URLs should be identified as primary and secondary if either:

There is a connection speed difference between the URLs (The faster connection listed as primary)	OR	One URL is only available when the other is down (primary URL being the most available)
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- The URLs should be listed as primary and alternate if:

The URLs have the same connection speed	AND	The URLs are customarily available simultaneously
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In the context of communication redundancy, a URL is considered available if all the TCP/IP facilities are properly functioning up to and including the HTTP service. This includes firewalls, DNS servers, routers, hubs, LANs, etc. between HTTP Server's and Internet Service Provider's point of presence.

In this context redundancy refers to normal operations redundancy, not to disaster recovery contingencies. Disaster recovery contingencies are not addressed in NAESB Internet ET standards.

Private network connections to access NAESB Internet ET sites may be at any point on a party's firewall boundary at the party's discretion on a non-discriminatory basis. The specific type and speed of their connection should be mutually agreed. It is at the discretion of the party how multiple private network connections should be managed.

TCP Communications

NAESB Cybersecurity Related Standards 12.3.21 and 12.3.22 restrict the TCP ports used as a standard for Internet ET communications. The use of NAESB standard TCP ports may require modifications in the Sender's and Receiver's firewalls to allow for communications with various Trading Partners' Internet ET implementations. Parties should indicate to their Trading Partners which specific TCP ports are required to be opened to conduct electronic communication.

Appendix C of this manual provides the allowed list of TCP Ports.

Other Communication Protocols

HTTP POST - HTTP POST is the standard method for transporting Internet ET packages to Trading Partners. The POST method allows the upload of complete datasets without special encoding.

MIME 'multi-part' - Internet ET packages are created using the 'multi-part' content type.

Sending Internet ET Packages

Internet ET supports both Interactive Browser and Batch Browser. Interactive Browsers provide for low-cost access to Internet ET capabilities. A Batch Browser allows organizations to maximize their level of automation. The Batch Browser can be an event-driven mechanism used to push Internet ET packages to Trading Partners in real-time or near real-time, while providing better audit trails.

Receiving Internet ET Packages

Receiving Internet ET packages and transaction payloads requires a Receiving Program. The Receiving Program:

- Parses the Internet ET package parameters and files to determine if the appropriate parameters were transmitted,
- Saves a log including a timestamp for the package.
- Stores the payload file.
- Sends the Receipt as an HTTP Response to the Sender/Client with the timestamp, and other required Receipt elements.

In some cases, the Receiving Program decrypts the file prior to sending the Receipt. In this scenario, decryption errors would be communicated in the Receipt. Some Trading Partners decrypt after sending the Receipt. Decryption errors detected after the Receipt is sent are communicated to Trading Partners using Internet ET Error Notification standards. Parties should notify Trading Partners of how decryption errors will be communicated.

After the Receiving Program performs its functions without errors, the payload file is forwarded to other processes including security, translation, and back-office systems.

Security

NAESB Internet ET establishes several security measures as standards to ensure a minimum level of confidence in conducting business over the Internet, and to provide uniformity in the implementation of security. Four security concepts, often referred to by the acronym PAIN, are vital to protecting Internet ET packages:

- **Data Privacy**
- **Authentication**
- **Data Integrity**
- **Non-repudiation**

Data Privacy and Encryption

Privacy is the assurance to an entity that no one can read a particular piece of data except the receiver(s) explicitly intended. Data privacy is accomplished by encrypting Payload Files. Internet ET allows encryption using both OpenPGP and PGP, as defined in Appendix A of this manual.

Internet ET uses base64-encoding and Transport Layer Security to protect username and password.

Authentication

Authentication is the assurance to one entity that another entity is who he/she/it claims to be. HTTP Basic Authentication is the required standard to prevent intruders from connecting to Internet ET Web sites. Internet ET uses Transport Layer Security to protect usernames and passwords used in authentication. Optional techniques such as firewall security enable further authentication.

Integrity

Integrity is the assurance to an entity that data has not been altered, intentionally or unintentionally, between there and here, or between then and now. Data Integrity is established via OpenPGP/PGP encryption, and via the 'content-length' HTTP header field.

Non-Repudiation

Non-repudiation is the assurance to an entity that a party cannot deny having engaged in the transaction, or having sent the electronic message. It is like a Notary seal. The Sender of a file will include in the Internet ET package a digital signature that is created using their Private Key. The Receiver knows the Sender is legitimate by decoding the digital signature using the Sender's Public Key.

B. GENERAL STANDARDS

Principles:

- 0.1.1 An entity is a person or organization with sufficient legal standing to enter into a contract or arrangement with another such person or organization (as such legal standing may be determined by those parties) for the purpose of conducting and/or coordinating energy transactions.
- 0.1.2 For NAESB WGQ purposes, there should be a unique entity common code for each entity name and there should be a unique entity name for each entity common code.

Standards:

- 0.3.1 Entity common codes should be 'legal entities', that is, Ultimate Location, Headquarters Location, and/or Single Location (in Dun & Bradstreet Corporation ('D&B') terms). However, in the following situations, a Branch Location (in D&B terms) can also be an entity common code:
 - 5. when contracting party provides a D-U-N-S® Number at the Branch Location level;
 - OR**
 - 6. to accommodate accounting for an entity that is identified at the Branch Location level.

C. INTERNET ELECTRONIC TRANSPORT RELATED STANDARDS

Principles:

- 12.1.1 The Internet Electronic Transport (ET) does not pick winners, rather it should create an environment where the marketplace can dictate a winner or winners.
- 12.1.2 Internet ET solutions should be cost effective, simple and economical.
- 12.1.3 Internet ET solutions should provide for a seamless marketplace for energy.
- 12.1.4 Parties should interface with Third-Party vendors according to NAESB Internet ET standards.
- 12.1.5 Electronic communications between parties to the transaction should be done on a non-discriminatory basis, whether through an agent or directly with any party to the transaction.
- 12.1.6 Protocols and tools that parties elect to support should be 'Internet-compatible'.
- 12.1.7 Trading Partners should maintain redundant connections to the public Internet for NAESB Internet ET Web sites. These redundant connections should be topographically diverse (duality of) paths to minimize the probability of a single point of failure.

- 12.1.8 The Cybersecurity Related Standards manual should be reviewed and updated for technology changes by the WGQ EDM subcommittee, prior to the publication of the WGQ Standards manual. This review should be accomplished in sufficient time for review by the respective Executive Committee prior to publication.
- 12.1.9 Transportation Service Provider Internet ET implementations should minimize the use of outbound ports required to be opened on the Client-side firewall.

Definitions:

- 12.2.1 'Internet ET Testing'. Testing Electronic Packages between Trading Partners includes testing of: A) Connectivity; B) Encryption/Decryption; and C) Digital signatures.
- 12.2.2 'Fail-over' defines a prescribed process executed when a NAESB Internet ET Client fails to establish a connection to the target NAESB Internet ET Server.
- 12.2.3 'Trading Partner' is a party that enters into an agreement with another party to transact business electronically using the Internet ET standard.
- 12.2.4 'Originating party' is any party originating/creating the package. This could also include a Third-Party.
- 12.2.5 'Third-Party' is any organization that a trading party uses to provide services to comply with the required elements of the Internet ET.
- 12.2.6 'Receiving Party' is any party that hosts (either in-house or outsourced) an Internet ET compliant server capable of receiving Internet ET packages.
- 12.2.7 'Receiving Program' is a program or set of programs that process HTTP Requests from a Sender. The Receiving Program is responsible for generating the 'gisb-acknowledge-receipt', which includes any party that hosts (either in-house or outsourced) an Internet ET compliant server capable of receiving Internet ET packages.
- 12.2.8 'Trading Partner Agreement', or 'TPA' is a legal binding agreement between trading parties that defines each party's expectations and responsibilities. This agreement often dictates service level agreements and problem remediation processes.
- 12.2.9 'Batch Browser'. A Browser that can be run with little or no manual operation or intervention. See 'Browser'.
- 12.2.10 'Browser'. A software program capable of generating HTTP Requests, including HTTP POST requests.
- 12.2.11 'Client'. The computer hardware and software used by the Sender to transmit an Electronic Package to the Receiver's Server. A Client can be fully-automated or manual.
- 12.2.12 'COTS'. Commercial Off-The-Shelf; software that can be purchased and that requires little or no customization.
- 12.2.13 'Electronic Package'. A data stream sent via HTTP POST that contains envelope header information and Payload File(s). The Payload Files are encrypted using defined Internet ET encryption techniques.
- 12.2.14 'Error Notification'. Error Notification is a package sent from the Receiver of the original data to the Sender when errors are trapped after the Internet ET Receipt is sent, for example, when decryption errors are detected after the Internet ET Receipt has been sent.

- 12.2.15 'HTTP Request'. The stream of data sent from the Client to the Server that includes header information and payload data.
- 12.2.16 'HTTP Response'. The stream of data sent from the Server to the Client in response to an HTTP Request, including the Receipt.
- 12.2.17 'HTTP Server'. The computer hardware and software used by the Receiver to receive HTTP Requests from the Sender's Client, and to send HTTP Responses to the Sender's Client. The Server is an HTTP/Web Server.
- 12.2.18 'IETF'. Internet Engineering Task Force; a body of technical experts that set standards for the Internet known as Request for Comments (RFC's).
- 12.2.19 'Interactive Browser'. A Browser that requires manual operation or intervention. See 'Browser'.
- 12.2.20 [Deleted]
- 12.2.21 'Internet ET' or 'Internet Electronic Transport'. The NAESB standards for the secure transport of electronic information between Trading Partners, building upon WGQ EDM Version 1.7.
- 12.2.22 'Payload Files'. The data contents inside of an Electronic Package. NAESB Internet ET is content-independent.
- 12.2.23 'Protocol Failure'. A Protocol Failure occurs any time a sending party's NAESB Internet ET server cannot connect to the Receiving Party's NAESB Internet ET server. For example, if a server tries to connect to a server and fails, or tries to post a file and fails, this is a Protocol Failure.
- 12.2.24 'Exchange Failure'. An Exchange Failure is when a sending party's NAESB Internet ET server has had three or more Protocol Failures within a 30 minute period.
- 12.2.25 'QEDM'. Quadrant Electronic Delivery Mechanism; the set of standards defining WGQ EDM standards for EDI, flat-files, electronic bulletin boards, and other technologies. The QEDM manual excludes electronic transport and cybersecurity practices and standards.
- 12.2.26 'Receipt'. The HTTP Response sent from the Receiver to the Sender that includes the 'gisb-acknowledge-receipt' section with a timestamp and OK/error status.
- 12.2.27 'Receiver'. The party that receives an Internet ET Electronic Package.
- 12.2.28 'Sender'. The party that sends an Electronic Package.
- 12.2.29 'QoS'. Quality of Service; term used to define what level of network bandwidth is guaranteed or assured. The Internet does not offer guaranteed quality of service.
- 12.2.30 'Trading Partner Worksheet' or 'TPW'. is a document containing detailed information related to the technical implementation of the Internet Electronic Transport. This document is a technical supplement to the Trading Partner Agreement (TPA) and provides information which may change periodically without the need to renegotiate the TPA.
- 12.2.31 'TCP'. Transmission Control Protocol; IETF RFCs 793, 1122, 1323
See <http://www.itprc.com/tcpipfaq/default.htm>.
- 12.2.32 'RSA'. A mathematical algorithm for encryption developed by Rivest/Shamir/Adleman.
See <http://world.std.com/~franl/crypto/rsa-guts.html>.
- 12.2.33 'TLS'. Transport Layer Security; a privacy technique that uses encryption to hide information from electronic observers on the Internet.

- 12.2.34 'PGP'. Pretty Good Privacy; software used to create Public and Private Keys for privacy and digital signature applications.
- 12.2.35 'Private Key'. The sequence of digits known as a 'key' that is kept private by the owner of a digital certificate, and is used by the certificate owner in encryption and decryption algorithms.
- 12.2.36 'Public Key'. The sequence of digits known as a 'key' that an owner of a digital certificate shares with Trading Partners. The Trading Partners use the public key in encryption and decryption algorithms in electronic transactions with the certificate owner.
- 12.2.37 'HTTP'. Hypertext transport protocol; Assumes version HTTP/1.1; IETF RFCs 2616, 2069. See <http://www.w3.org/Protocols/Specs.html>.
- 12.2.38 'MIME'. Multipurpose Internet Mail Extensions; IETF RFCs 2045, 2046, 2047, 2048, 2049; See <http://www.faqs.org/rfcs/rfc2045.html>.
- 12.2.39 'OpenPGP'. Open Pretty Good Privacy; open source encryption used to create Public and Private Keys for privacy and digital signature applications.
- 12.2.40 'Internet Flat File' is the term used within NAESB WGQ FF/EDM to describe the transfer of flat files using an Interactive Browser.
- 12.2.41 Testing data sets between Trading Partners includes testing of:
1. intended business results,
 2. Internet ET, and
 3. related EDI/EDM and, where supported, FF/EDM implementation issues.
- Testing should include enveloping, security, data validity, and standards compliance (e.g. ANSI X12 and NAESB WGQ QEDM Related Standards).

Standards:

- 12.3.1 All parties sending and receiving data should accept a TCP/IP connection.
- 12.3.2 The designated Internet ET Server/Receiver site should be accessible via the public Internet. This does not preclude location of the designated site on a private intranet, as long as the designated site is also accessible via the public Internet.
- 12.3.3 All Trading Partners should communicate using the minimum HTTP version, as specified in Appendix A of the WGQ Cybersecurity Related Standards.
- 12.3.4 A timestamp designates the time a file is received at the Receiver's designated site. The timestamp consists of the 'time-c' data element.
- 12.3.5 The Receiver generates a timestamp upon the successful receipt of a complete file. The timestamp should be generated by the Receiving Program immediately, prior to further processing by the Receiving Program.
- 12.3.6 After timestamp generation, the Receiver sends an immediate HTTP Response to the Sender. The 'gisb-acknowledgement-receipt', which includes the timestamp data element(s), is the primary part of the HTTP Response.
- 12.3.7 The Server clock generating the timestamp should be synchronized with the National

Institute of Standards and Technology (NIST) time in order to mitigate discrepancies between the clocks of the Sender and Receiver. Computer clocks should be synchronized as necessary to ensure at minimum +/- 5 second synchronization with an atomic clock. Specific business processes may have tighter synchronization requirements.

- 12.3.8 The HTTP Response should be sent to the Internet Protocol (IP) address of the HTTP Request.
- 12.3.9 At a minimum, one designated site for receipt should be identified for each Trading Partner. That site should be identified by a specific Uniform Resource Locator (URL). This does not preclude multiple designated sites being mutually agreed to between Trading Partners.
- 12.3.10 The Sender should make three attempts to complete a unit of work. A unit of work consists of one complete HTTP POST transaction as defined in the technical specification of the HTTP protocol (IETF RFC 1945).
- 12.3.11 A failure to complete a unit of work is a Protocol Failure.
- 12.3.12 The Internet ET roles for Sender and Receiver are defined in the following table. The entire table defines a unit of work:

Client (Sender)	Server (Receiver)	Receiving Program (Receiver)
	Listen for Connect	
Connect	Accept Connection	
Write HTTP Request	Read HTTP Request	Start of Receipt
Write HTTP Request	Read HTTP Request	
EOF (send)	Read HTTP Request	End of Receipt
Read HTTP Response	Write HTTP Response	
Received		
EOF HTTP Response		

- 12.3.13 Trading Partners should implement all security features (privacy, secure authentication, integrity, and non-repudiation) using a file-based approach using an Open PGP product, or, on a mutually agreed basis, PGP, as specified in Appendix A in the WGQ Cybersecurity Related Standards.
- 12.3.14 Trading Partners should implement HTTP Basic Authentication using Transport Layer Security, as specified in Appendix A in the WGQ Cybersecurity Related Standards.
- 12.3.15 Encryption keys should be self-certified. The exchange of Public keys should be completed electronically such as via email. Key policies, including key exchange policies should be communicated to Trading Partners.
- 12.3.16 Encryption keys should have a limited lifetime whose duration is determined by the key's owner. A key's end of life is expressed in the expiration date field contained in each Public Key. A lifetime of one year or less is recommended.
- 12.3.17 Internet protocols should be used for accessing all industry business functions.
- 12.3.18 Batch and Interactive Browsers should use Internet-compatible common browser software.
- 12.3.19 Trading Partners should use common codes for legal entities for the Internet ET 'to' and 'from' data elements.
- 12.3.20 Private network connections to NAESB Internet ET servers, which include all NAESB Internet ET standardized Internet communication, may be at any point on a party's

firewall boundary at the party's discretion on a non-discriminatory access basis. The specific type and speed of these connections should be mutually agreed. It is at the discretion of each party on how multiple private network connections should be managed, so long as such management is done on a non-discriminatory access basis.

- 12.3.21 Parties should be limited to the NAESB Internet ET approved list of available TCP ports for Internet ET implementations.
- 12.3.22 Internet ET implementations should not require any inbound ports to be opened on the Sender's firewall.
- 12.3.23 Internet ET servers should use Transport Layer Security, as specified in Appendix A in the WGQ Cybersecurity Related Standards.
- 12.3.24 A decryption response should be sent no later than 60 minutes from the initial payload receipt, or within a timeframe as specified in the WGQ QEDM standards.
- 12.3.25 The information contained in the Trading Partner Worksheet and Trading Partner Agreement as well as Trading Partner digital signature and encryption keys should be secured and considered 'company proprietary' or 'company confidential' information.
- 12.3.26 The North American Energy Standards Board Wholesale Gas Quadrant should not set standards for site-level security. Individual organization security standards should be relied upon.

Individual organizations should address, at a minimum, the following items as related to NAESB WGQ EBB/EDM, EDI/EDM, and/or FF/EDM:

- As software or hardware vendors release changes (i.e. release, patch) to their product that address potential security weaknesses and major security release associated bulletins, Trading Partners should evaluate the proposed changes within 30 days. If the Trading Partner identifies a security risk, the change should be implemented as soon as reasonably practicable based on the severity of the risk and the compatibility with the specific computing environment.
- If vulnerabilities that directly impact the security and integrity of data transmission between Trading Partners cannot be addressed as soon as reasonably practicable, based on the severity of the risk, the risk should be communicated to the affected Trading Partners.
- Trading Partners should implement controls to (1) proactively evaluate their computer hardware and software to identify known security risks; (2) detect malicious activity; and (3) log information for forensic analysis.
- Trading Partners should have a business continuity plan in the event of critical systems' failures.

- 12.3.27 Trading Partners should mutually agree to use either the current, or the immediately previous, published version of the NAESB WGQ Cybersecurity Related standards as a basis under which to operate, unless: 1) Security standards with the current NAESB Internet ET and/or the WGQ QEDM are not compatible with security standards in the previous version, and/or 2) specified otherwise by an applicable regulatory authority.
- 12.3.28 Access to Internet Flat File EDM should be protected by HTTP Basic Authentication using Transport Layer Security, as specified in Appendix C of the WGQ Cybersecurity Related Standards.
- 12.3.29 Access to the Customer Activities Web Site should be protected by HTTP Basic Authentication using Transport Layer Security, as specified in Appendix B of the WGQ Cybersecurity Related Standards. Unless specified otherwise by an applicable regulatory authority, a Customer Activities Web site should require either a single logon/password pair or Multi-Factor Authentication for each user session.
- 12.3.30 Custom downloadable modules presented by a Customer Activities Web site should be signed by the author. The signatures on these modules should be communicated in advance to Web site users.
- 12.3.31 Providers of Customer Activities and Informational Postings Web sites should ensure that the site operates within the guidelines of the “Minimum Technical Characteristics and Guidelines for the Developer and User of Customer Activities and Informational Postings Web Sites” described in the Appendix B of the NAESB WGQ Cybersecurity Related Standards.

D. INTERPRETATIONS

In Version 3.0, NAESB WGQ deleted the interpretations of standards that relate to this manual by giving clarity to existing manual textual language and samples and/or adding/changing/deleting Principles, Definitions and/or Standards language.

RELATED STANDARDS

COMMON CODES

Internet ET uses the D-U-N-S® Number as the common company identifier for the HTTP Request and Response data dictionary 'to' and 'from' HTTP header elements. The D-U-N-S® Number is a 9-digit number assigned to companies by the Dun & Bradstreet Corporation (D&B). The D-U-N-S+4® Number is a 10- to 13-digit number, where characters 10 through 13 are arbitrarily assigned by the owner of the D-U-N-S® Number.

For Internet ET Common Code purposes, an entity will use one and only one D-U-N-S® Number. Entity common codes should be 'legal entities,' that is, Ultimate Location, Headquarters Location, and/or Single Location (in D&B terms). However, in the following situations, a Branch Location (in D&B terms) can also be an entity common code:

1. When the contracting party provides a D-U-N-S® Number at the Branch Location level.
2. To accommodate accounting for an entity that is identified at the Branch Location level.

Since D&B offers customers the option of carrying more than one D-U-N-S® Number per entity, please refer to NAESB's Web Page at www.naesb.org for directions on determining the one and only one D-U-N-S® Number constituting the NAESB Internet ET Entity Common Code.

QUADRANT ELECTRONIC DELIVERY MECHANISM (QEDM)

In NAESB business processes, the Internet ET standards are used in conjunction with the WGQ Quadrant Electronic Delivery Mechanism standards. These standards include, but are not limited to, X12/EDI standards, flat-file standards, web standards, etc.

PARTY ROLES

Various types of parties are involved in NAESB business processes and the use of Internet ET, including distribution companies, end-users, regulatory entities, service providers, and suppliers.

TECHNICAL IMPLEMENTATION - INTERNET ET

INTERNET ET TECHNOLOGIES

The NAESB Internet ET uses the following technologies and components to securely and reliably transport Electronic Packages to Trading Partners:

- OpenPGP and PGP encryption and digital signatures
- TCP/IP and HTTP POST. Internet ET uses a specifically-structured HTTP POST to transport payload data from one Trading Partner to the other
- MIME multi-part encoding. Internet ET package structure requires that each section of the package be encoded
- A 'Client', running at the Sender's site as 'Batch' or 'Interactive' Browser software. This software is referred to in this document as 'Client'
- A 'Server' running at the Receiver's site, usually on a dedicated computer. This is a Web or HTTP Server, and is referred to in this document as 'Server'

ELECTRONIC TRANSPORT LIFE CYCLE

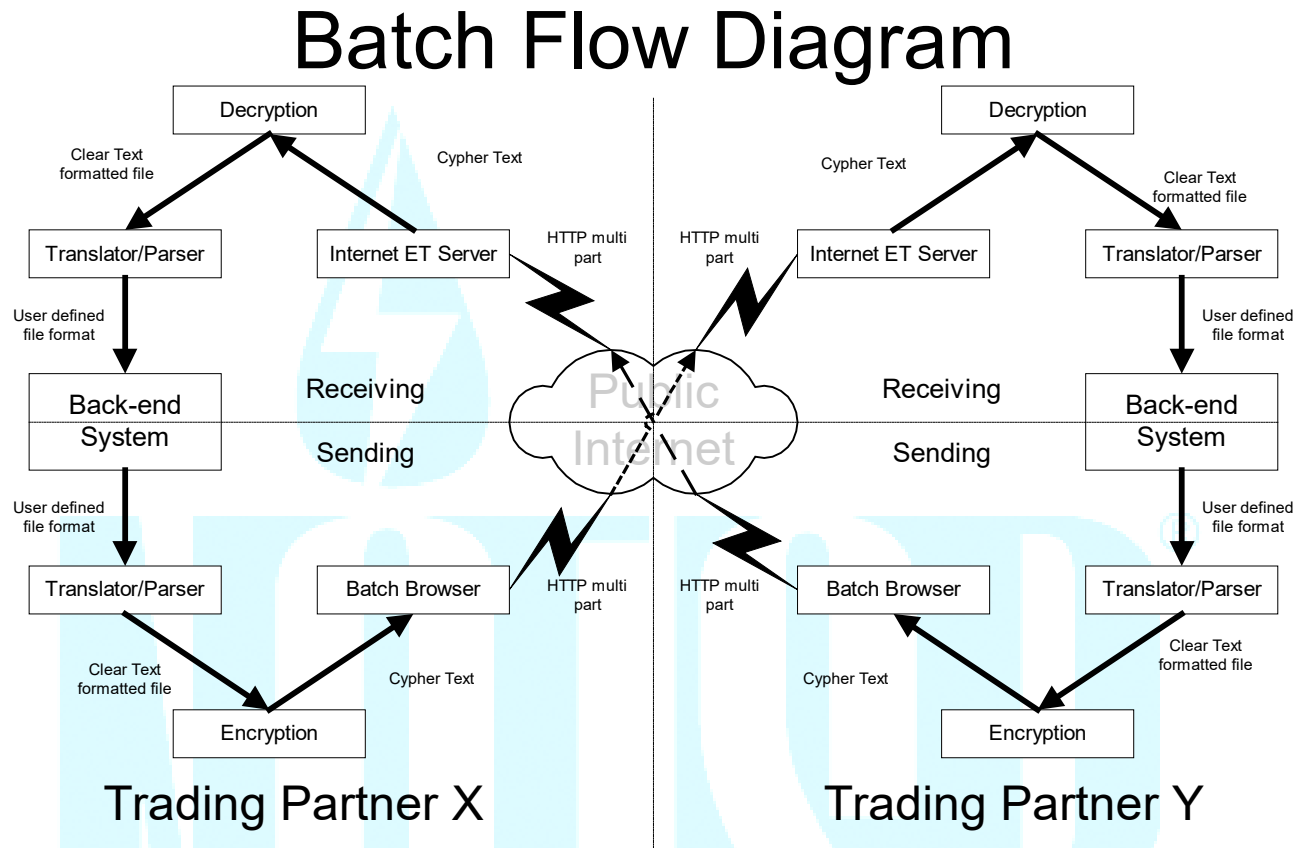
The life cycle of an Electronic Package using Internet ET is described below:

Sender		Receiver
• Collects payload data to be sent • Encrypts payload • Prepares digital signature		
• Uses Browser to create Electronic Package multi-part HTTP Request with header data elements and payload. • Uses HTTP POST to send the Electronic Package to the Receiver	➔	
		• Receives the HTTP Request on their Web/HTTP Server • Validates Sender information from HTTP Request Header data elements and payload • **Decrypts payload file • Prepares Receipt • Checks digital signatures
	↙	• Sends Receipt with either 'OK' or error message
• Updates logs • If errors, correct errors then repeat process		• **Decrypts payload file
	↙	• If errors in decryption, sends Error Notification to Sender
• Receives Error Notification • Updates logs • correct errors then repeat process		• Updates logs • If no errors, Receiver processes contents of payload

**Parties may choose to decrypt file before or after Receipt is sent to Sender.

Batch Flow Diagram

The flow of data to and from Trading Partners in an automated environment is diagrammed below.



ANATOMY OF AN INTERNET ET PACKAGE

An Internet ET package consists of the following sections:

- **Envelope header.** This section contains the envelope information needed to communicate who the Sender and Receiver are, as well as other envelope information.
- **Payload.** This section contains the payload file. Internet ET allows for only one payload file per package.
- **Digital Signatures.** The package should contain a section that is the digital signature.

ENVELOPE DATA DICTIONARY

The data dictionary on the next page details standard data elements, each with element name and description.

Data Dictionary for Internet ET

Business Name	Definition	Format	Usage*	Condition
from**	the party sending the transaction	Common Code Identifier format	in Request; M	used in file transmittal; displayed in HTTP Response; and, used in posting back decryption-related errors
input-data	the filename for the transaction data set transmitted	including drive letter and directory name with filename if needed	in Request; M	Used in file transmittal of any transaction data sets; and, used for posting back all transaction value pairs for a transmittal that had decryption-related errors.
input-format	descriptor of the data format used for the file transmitted	X12;FF; error	in Request; M	NAESB standard format indicator used in file transmittal
receipt-disposition-to	the party to receive receipts, the value should be the same as the 'from'	Common Code Identifier format	in Request; M	used in file transmittal and in posting Error Notifications
receipt-report-type	type of receipt type being requested by Sender	gisb-acknowledgement-receipt	in Request; M	used in file transmittal and in posting Error Notifications
receipt-security-selection	used to request signed receipts	signed-receipt-protocol=required,pgp-signature;signed-receipt-micalg=required,md5	in Request; M	used in file transmittal and in posting Error Notifications
refnum	used by the party to assign a message identifier unique over all time for tracing purposes. For the Sender, this ID should not be duplicated for resends.	maximum 40-character integer value	in Request; MA	May be used by Sender to send tracking information to a recipient. Use of this data element is by mutually agreed. This data element is conceptually similar to a Message-ID filed within RFC 822.
refnum-orig	for original send, refnum-orig is identical to refnum. for resend, refnum-orig is the refnum of the original package.	maximum 40-character integer value	in Request; MA	Used in conjunction with refnum.

Business Name	Definition	Format	Usage*	Condition
request-status	status describing success or failure of transmission at recipient Server	ok; EEDM###:error description; WEDM###:warning description. see Table 1, 'Internet ET Standard Error Codes and Messages'	in Response; M	'ok' is returned if all is fine with processing; error messages/warnings and their related descriptions are returned if problems were encountered in processing.
server-id	uniquely identifies the Server processing the transaction	domainname or hostname.domainname; no embedded spaces allowed	in Response; M	displayed in the HTTP Response and posted back for any decryption-related errors
time-c	the time file transfer is complete at the Server	yyyymmddhhmmss	in Response; M	displayed in the HTTP Response and posted back for any decryption-related errors; refer to QEDM for quadrant-specific use
to **	the party the transaction was sent to	Common Code Identifier format	in Request; M	used in file transmittal and displayed in HTTP Response and posted back for any decryption-related errors
transaction-set	name of the document type being sent	8 character code; refer to NAESB REQ and WGQ Implementation Guides, Related Standards Tab, Hypertext Transfer Protocol (HTTP) section, HTTP transaction-set Code Values table.	in Request; MA	used in file transmittal
trans-id	sequential number assigned to the transaction by the Server upon processing before being passed to the decryption process	integer up to 15 characters in length	in Response; M	displayed in the HTTP Response and posted back for any decryption-related errors
version	the NAESB Internet ET version being used by the Sender	numeric, decimal notation (e.g. 1.6)	in Request; M	used in file transmittal and in posting Error Notifications

*The **Usage** column defines whether the element appears in the HTTP Request (Client-generated) or the HTTP Response (Server-generated), the order in which the element appears in the data stream, and whether the field is Mandatory (M) or Mutually-Agreed-To (MA).

** Common Code Identifier

SENDING INTERNET ET PACKAGES

General Flow

The following is an example of the steps necessary to send an Internet ET package:

1. Open HTTP connection.
2. Check connection status. If in error, re-queue package according to Internet ET standards. This check should be performed here and throughout the following processes.
3. Post, including a) Authentication, b) Send multipart form, c) Receive HTTP Response data.
4. Check connection status. If in error, re-queue package according to Internet ET standards.
5. Check HTTP status code (200 is good, less than 300 may be acceptable). If status is not successful, re-queue package according to Internet ET standards.
6. Close connection - wait for other end to close in a reasonable time.
7. Parse HTTP Response data elements.
8. If request-status ok, then log success.
9. If request-status error, then log error.
10. If no valid request-status, re-queue package according to Internet ET standards. [®]
11. Remove package from sending queue when successful or when failed completely.

The sending party must include the 'receipt-security-selection' data element in the posted data. The Receiving Party must digitally sign the 'gisb-acknowledgement-receipt' and encapsulate the 'gisb-acknowledgement-receipt' and digital signature body parts within a MIME envelope with a 'content-type' of 'application/pgp-signature'.

Use of Refnum and Refnum-orig

Refnum and Refnum-orig are mutually-agreed data elements; meaning, if one or the other Trading Partner doesn't agree to use these data elements, then the Refnum and Refnum-orig functionality will not be utilized in their data exchange.

The first time a package is sent the refnum and refnum-orig should be identical 40-digit or less integers. The refnum data element is always unique over time.

If a party does not receive the NAESB response, or a party does not receive an intelligible NAESB response, the package should be resent with a new refnum, and with the refnum-orig equal to the original refnum used in the initial transmittal of the package.

Refnum and Refnum-orig Example

Package Send	refnum	refnum-orig
First send	123467890123456	123467890123456
First resend	223467890123457	123467890123456
Second resend	323467890123458	123467890123456

- **Abuse of Refnum and Refnum-orig**

There is potential for abuse/error when using the functionality of these mutually-agreed data elements. Parties should monitor for the re-use of numbers used with both refnum and refnum-orig for the receipt and delivery functions.

Example of Abuse by re-using Refnum in “First resend”

Package Send	refnum	refnum-orig
First send	123467890123456	123467890123456
First resend	123467890123456	123467890123456

Example of Abuse by Sending Bogus Refnum-orig in “First resend”

Package Send	refnum	refnum-orig
First send	123467890123456	123467890123456
First resend	223467890123457	010101010101010

Example of Abuse by re-use Refnum and Refnum-orig in “New package”

Package Send	refnum	refnum-orig
First send	123467890123456	123467890123456
First resend	223467890123457	123467890123456
New package	123467890123456	123467890123456

Using an Interactive Browser for Internet ET

Electronic Packages can be uploaded to a Trading Partner using an Interactive Browser secured using Transport Layer Security. Sending Electronic Packages via an Interactive Browser is ideal for a small volume of package transfers, or as a back-up method to any batch or automated process.

To use an Interactive Browser to upload data, an HTML document must be created with an HTML <FORM> element that allows the Sender to type in any necessary data elements, such as ‘to’, ‘from’, ‘input-format’, and the name of the file to be uploaded. When the user submits the form, an HTTP POST is sent to the Server with the package, which includes the uploaded file and the required data elements.

The following example is an HTML document with a form that specifies the POST method and contains the required data elements. This type of HTML form could be used with any Browser that supports multipart POST with a file upload.

EXAMPLE: HTML DOCUMENT WITH A FORM FOR MULTIPART POST USING AN INTERACTIVE BROWSER:

```
<HTML><HEAD><TITLE>NAESB Internet ET Package Upload</TITLE><H1><CENTER>NAESB Internet ET
Package Upload</CENTER></H1></HEAD>
<BODY><HR>
<FORM ENCTYPE="multipart/form-data" ACTION="http://www.target.server/cgi-bin/upload.exe"
METHOD="POST">
Enter Common Code Identifier for 'From' and 'To':
From:
<INPUT TYPE="text" NAME="from" SIZE=20 VALUE=""><br>
To:
<INPUT TYPE="text" NAME="to" SIZE=20 VALUE=""><br>
```

```

NAESB Internet ET Version:
<INPUT TYPE="text" NAME="version" SIZE=5 VALUE="1.6"><br>
Deliver Receipt To:
<INPUT TYPE="text" NAME="report-disposition-to" SIZE=20 VALUE=""><br>
Receipt Type:
<INPUT TYPE="text" NAME="receipt-report-type" SIZE=30
VALUE="gisb-acknowledgement-receipt"><br>

IF requesting signed receipts also include: Receipt Type:
<INPUT TYPE="text" NAME="receipt-security-selection" SIZE=30
VALUE="signed-receipt-protocol=required, pgp-signature; signed-receipt-
micalg=required, md5"><br>
Format of this file:
<INPUT TYPE="text" NAME="input-format" SIZE=6 VALUE="X12"><br>
Send this file:
<INPUT NAME="input-data" TYPE="FILE"><br>
<INPUT TYPE="submit" VALUE="Send File"><br>
</FORM>
</BODY></HTML>

```

The important characteristics of the form within the HTML document are:

- ENCTYPE= specifies the encoding type. The 'multipart/form-data' encoding type is identified as the standard encoding methodology.
- ACTION= specifies the URL that will receive the uploaded data. The TPW identifies the URLs for both parties.
- METHOD= specifies the HTTP protocol method. 'POST' has been defined as the Internet ET standard method.
- <INPUT ...>. HTML INPUT elements include the required data elements such as 'from', 'to', and 'input-format'. Refer to the data dictionary for the complete list of required data elements.

When a user selects the 'Send File' button, the Interactive Browser will take the values entered in the input fields and reformat them into a data stream, formatted according to the encoding type. The file identified for upload is opened and its contents are included in the data stream. The data stream is then sent to the URL specified by '**ACTION=**', which indicates a Server Receiving script or program written to receive the package.

Using a Batch Browser for Internet ET

A Batch Browser is used by companies that want to automate their transport processes and/or prefer to minimize human involvement. A Batch Browser is initiated by a program or a script.

A Batch Browser can be created via custom programming. A Batch Browser is coded to perform the same formatting as an Interactive Browser, formatting a data stream that conforms to the HTTP and Internet ET protocols. A Batch Browser must be coded as a 'TCP sockets' program. See the section 'Writing a Batch Browser'.

Authentication

Userids and passwords must be base64-encoded. HTTP Basic Authentication includes a 'userid' and 'password', and is protected using Transport Layer Security. Interactive Browsers include a basic authentication feature that automatically prompts for 'userid' and 'password'. In a Batch Browser, the authentication must be specifically coded. The 'userid' and 'password' are to be base64-encoded within the document header. Base64-encoding utilities are readily available on the Internet as either public domain software or commercial libraries.

Server Response

The Server will send a 'gisb-acknowledgement-receipt' in the HTTP Response to the Client before dropping the Client's connection. The Server applies a digital signature to the 'gisb-acknowledgement-receipt' and encapsulates the entire package in a MIME envelope of 'content-type: application/pgp-signature'.

The 'gisb-acknowledgement-receipt' returned from the Server contains the 'time-c' Receipt timestamp that is recorded when the final byte from the package upload is received and stored. This Receipt timestamp is the official timestamp regarding transaction turnaround deadlines as defined in Internet ET and WGQ QEDM standards. This timestamp and all other pertinent package transmittal information should be logged by the Receiver when the posted package is stored on the Server and logged by the Client. Errors or warnings should be logged at both the Client and Server.

Sender HTTP Request Data Elements

The HTTP Request will provide all required data elements in the ORDER DEFINED BELOW. Any 'mutually-agreed-upon' data elements will follow the required data elements in the data stream. Refer to the section 'Data Dictionary for Internet ET' for descriptions of these data elements.

Required Data Elements, Listed in the Required Order:

1. from
2. to
3. version
4. receipt-disposition-to
5. receipt-report-type
6. input-format
7. input-data
8. receipt-security-selection

Mutually Agreed Upon Data Elements

9. transaction-set
10. refnum
11. refnum-orig

Writing a Batch Browser

A Batch Browser Client needs to simulate the actions of an Interactive Browser Client. As stated earlier, the Interactive Browser Client will take the HTML form, reformat the information according to the HTTP protocol, then send the data stream to the Server. The reformatting adds a header and places field delimiters around the data items.

A Batch Browser needs to produce the same kind of data stream and, therefore, writing a Batch Browser requires some specific knowledge of the HTTP protocol.

EXAMPLE: A TYPICAL HEADER SENT TO THE SERVER

```
POST /cgi-bin/AS2dispatcher HTTP/1.1
Referer: http://www.get.a.life/upl.htm
Connection: Keep-Alive
User-Agent: brow v0.1 XYZ Corp.
Host: localhost
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, */*
content-type: multipart/form-data; boundary=-----87453838942833
Content-Length: 5379
```

POST Line - In the example above, the first line indicates the POST method was used and identifies which Receiving Program to call:

```
POST /cgi-bin/AS2dispatcher HTTP/1.1
```

Content Type - The 'content-type' line indicates that the encoding method is multipart, and identifies the character string used as the boundary.

```
content-type: multipart/form-data; boundary=-----87453838942833
```

Boundary String - The 'boundary=' identifies the string that will appear between each field as a delimiter. In this example, the boundary is comprised of 27 hyphen characters followed by a number.

```
content-type: multipart/form-data; boundary=-----87453838942833
```

The boundary can be ANY character string that ISN'T USED ANYWHERE ELSE IN THE PACKAGE BEING SENT. Most implementations accomplish this by using either the system clock or a random number so the string would not appear in any re-transmission(s) of the file. It is strongly recommended to use a relatively long string as a boundary.

The boundary string, when used as a separator, REQUIRES TWO HYPHEN CHARACTERS APPENDED TO THE FRONT of the string. The LAST boundary required in the form is TWO HYPHEN CHARACTERS APPENDED TO THE BACK of the separator boundary, used to indicate to the Server program that this is the end of the data.

```
-----87453838942833--
```

Content Length - The 'content-length' value should match the number of bytes contained in the entity body including the characters in the boundary lines, variable content, blank lines, etc. 'content-length' indicates to the Server how much data are going to come after this point. In the example above, the content length is:

```
Content-Length: 5379
```

Envelope / Required Data Elements. The envelope information for the package ('to', 'from', etc) is included in a series of boundaries that include the 'content-disposition' and 'name=' qualifiers, followed by the data element value. The example below includes the 'from' field as '123456789' and the 'to' field as '234567890'.

```
-----87453838942833
content-disposition: form-data; name="from"

123456789
-----87453838942833
content-disposition: form-data; name="to"

234567890
```

The 'content-disposition' identifier defines that 'form-data' is contained in the element. The 'name=' identifier defines the name of the data element. These data element names must match the name specified by Internet ET Data Dictionary. The 'name=' identifier is not completely relevant since the fields should be present in the correct order, but this field should be checked to verify the validity of the form content.

The actual data value of the field is always preceded by a blank line. This is typically used as a marker for the Server program to indicate that a data value will follow. For example, note the blank line preceding 'X12' in the example. In most programming libraries and commercial products the starting delimiter is '\r\n\r\n' (C notation).

```
-----87453838942833
content-disposition: form-data; name="version"

1.9
-----87453838942833
content-disposition: form-data; name="receipt-disposition-to"

123456789
-----87453838942833
content-disposition: form-data; name="receipt-report-type"

gisb-acknowledgement-receipt
-----87453838942833
content-disposition: form-data; name="input-format"

x12
```

Payload. The content or ‘payload’ (EDI, etc) is encrypted and included in its own boundary section.

The data field containing the Internet ET payload file has two extra identifiers. The ‘filename=’ element indicates the name of the file sent from by the Sender. In the example the name of the file is ‘c:\temp\smallnom.bin’.

```
content-disposition: form-data; name="input-data"; filename="c:\temp\smallnom.bin"
```

The ‘content-type’ element indicates the type of the data being transmitted according to accepted Internet standards.

```
content-type: multipart/encrypted; boundary=--boundary2--200309090001;
protocol="application/pgp-encrypted"
```

Note that encrypted files can be multipart also, which means they will have their own boundary string.

```
-----87453838942833
content-disposition: form-data; name="input-data"; filename="c:\temp\smallnom.bin"
content-type: multipart/encrypted; boundary=--boundary2--200309090001;
protocol="application/pgp-encrypted"

----boundary2--200309090001
content-type: application/pgp-encrypted

Version: 1

----boundary2--200309090001
content-type: application/octet-stream

-----BEGIN PGP MESSAGE-----
Version: PGP 9.0

hQCMaZRGlpEOIOvdAQP+JMr0m/9+8yOL60Z9Vr6fFV81FCExB/o0xmwiMkiwYsHsz0e8sb7ErC340MrNA/dw3taGM
jmI+CXyRF/PLEdglNZE1ZCtNeL4YdIHAMLWwODG1QxhSucz8rMSgQ5mZzc0JwBdWLW70efgsu/9U1juJjYc1uZ6C0
3eFQv/43fkB+alAtTgydxX4g8QK664ad+Jo/XUICSmWBL66fqJR1KLeLf4wTaqGy174Aq48WpwwglEh785zC03UAw
0qg0ugMt86dPeyd91e2JigqwDYEf/DYEKD0J9BGiGpS/uAupNKj8Ocp2IWClxKOGUbxpVNOntqWHS/GntegvDE/7
/ewCxDxsnmQS95p0l141QZ1RqbeNaqx2Dq/ra9g65HNchOCzjul5Vi8HHf6Yhg2WnROe+npByyCue6rihgqNVOJwj
0Cvzpb4JE+gMDf3q4ISub1Fv7/+SSFHDdnhdC5YTpqf1Bc3B07hiLmtTXqNit31EbX9UVEl0bzSa9ZhxbC6/eS17N
uf5ZTDsh9nrk+QQJ6FeC9W4cqXLj7IZySaRO8Vtff+4ktqehYusT4kSpnk027aw4O/5jomUkfb22CAe4=
=Oiuo
-----END PGP MESSAGE-----
----boundary2--200309090001--
```

Boundary String Terminators - Each multipart stream must be terminated with the boundary string terminator. After the contents of the last data field, the boundary string and the required two-hyphen terminator indicate the end of the multipart encrypted payload. A second boundary terminator string indicates the end of the package:

```
----boundary2--200309090001--
-----87453838942833--
```

EXAMPLE: AN X12 EDI FILE ENCRYPTED WITH PGP

```
content-type: multipart/encrypted; boundary=--boundary2--200309090001;
protocol="application/pgp-encrypted"

----boundary2--200309090001
content-type: application/pgp-encrypted

Version: 1

----boundary2--200309090001
content-type: application/octet-stream

-----BEGIN PGP MESSAGE-----
Version: PGP 9.0

hQCMaZRG1pEOIOvdAQP+JMr0m/9+8yOL60Z9Vr6fFV81FCExB/o0xmwiMkiwYsHsz0e8sb7Er340MrNA/dw3taGMj
mI+CXyRF/PLEdg1NZE1ZCtNeL4YdIHAMLWwODG1QxhSucz8rMSgQ5mZzcOJwBdWLW70efgsu/9U1juJjYc1uZ6C03
eFQv/43fkB+alATtgYdxX4g8QK664ad+Jo/XUICSmWBL66fqJR1KLeLf4wTaqGy174Aq48Wpwwg1Eh785zC03UAW0
qg0ugMt86dPeYd91e2JigqWdYef/DYEKD0J9BGiGpS/uAupNKj80cp2IwClxKOGUbxpVNonNTqWHS/GntegvDE/7/
ewCxDxsnmQS95p01141QZ1RQbeNaqx2Dq/ra9g65HNchOCzjul5Vi8HHf6Yhg2WnROe+npByyCue6rihqqNVOJwj0
cVzpb4JE+gMDf3q4ISUb1Fv7/+SSFHDdnhdC5YTpqf1Bc3B07hiLmtTXqNit31EbX9.UVE1ObzSa9zhxbC6/es17N
uf5ZTDsh9nrk+QQJ6FeC9W4cqXLj7IZySaR08Vtff+4ktqeuHusT4kSpnk027aw40/5jomUkfb22CAe4=
=Oiuo
-----END PGP MESSAGE-----
----boundary2--200309090001--
```

EXAMPLE: AN X12 EDI DATA STREAM BEFORE ENCRYPTION:

```
content-type: application/EDI-X12

ISA~00~ ~01~AAA6300300~14~1234567890000 ~14~2345678900000
... more data from the X12 file...
IEA~1~000003616
```

EXAMPLE: A COMPLETE ELECTRONIC PACKAGE DATA STREAM

```

POST /cgi-bin/AS2dispatcher HTTP/1.1
Referer: http://www.get.a.life/upl.htm
Connection: Keep-Alive
User-Agent: brow v0.1 XYZ Corp.
Host: localhost
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, */*
content-type: multipart/form-data; boundary=-----87453838942833
Content-Length: 5379

-----87453838942833
content-disposition: form-data; name="from"

123456789
-----87453838942833
content-disposition: form-data; name="to"

234567890
-----87453838942833
content-disposition: form-data; name="version"

1.9
-----87453838942833
content-disposition: form-data; name="receipt-disposition-to"

123456789
-----87453838942833
content-disposition: form-data; name="receipt-report-type"

gisb-acknowledgement-receipt
-----87453838942833
content-disposition: form-data; name="input-format"

X12
-----87453838942833
content-disposition: form-data; name="input-data"; filename="c:\temp\smallnom.bin"
content-type: multipart/encrypted; boundary=--boundary2--200309090001;
protocol="application/pgp-encrypted"

----boundary2--200309090001
content-type: application/pgp-encrypted

Version: 1

----boundary2--200309090001
content-type: application/octet-stream

-----BEGIN PGP MESSAGE-----
Version: PGP 9.0

hQCMAzRG1pEOIOvdAQp+JMr0m/9+8yOL60Z9Vr6fFV81FCExB/o0xmwiMkiwYsHsz0e8sb7ErC340MrNA/dw3taGM
jmI+CXYRF/PLEdglNZE1ZCtNeL4YdIHAMLWwODG1QxhSucz8rMSgQ5mZzc0JwBdWLW70efgsu/9U1juJjYcluz6C0
3eFQv/43fkB+alATtgydxX4g8QK664ad+Jo/XUICSmWBL66fqJR1KLeLf4wTaqGy174Aq48Wpwwg1Eh785zC03UAW
0qg0ugMt86dPeyd91e2JigqwDYef/DYEKD0J9BGiGpS/uAupNKj8Ocp2IWClxKOGUbxpVNOntqWHS/GntegvDE/7
/ewCxDxsnmQS95p0l141QZ1RqbeNaqx2Dq/ra9g65HNchOCzjul5Vi8HHf6Yhg2WnROe+npByyCue6rihgqNVOJwj
0cVzpb4JE+gMDf3q4ISub1Fv7/+SSFHDDnhdC5YTpqf1Bc3B07hiLmtTXqNit31EbX9UVE10bzSa9ZhxbC6/es17N
uf5ZTDsh9nrk+QQJ6FeC9W4cqXLj7IZySaRO8Vtff+4ktqeuHusT4kSpnk027aw40/5jomUkfb22CAe4=
=Oiuo
-----END PGP MESSAGE-----
----boundary2--200309090001--
-----87453838942833--

```

RECEIVING INTERNET ET PACKAGES

General Flow

The following is an example of the steps necessary to receive an Internet ET package:

1. Parse multi-part form.
2. Validate HTTP Request data elements.
3. If HTTP Request data elements in error, return appropriate Internet ET standard error code in the HTTP Response data elements.
4. Save data.
5. Create 'gisb-acknowledgement-receipt'.
6. Produce a digital signature over the 'gisb-acknowledgement-receipt' created in step 5.
7. Encapsulate the 'gisb-acknowledgement-receipt' and digital signature body parts in a 'Content-Type' of 'multipart/signed envelope'.
8. Return HTTP Response with the 'gisb-acknowledgement-receipt' object back to Client.
9. Close connection.
10. Log final results.
11. Route data file to the next process based upon 'input-format'.

Overview of Web Server Receiving Programs

The HTTP Server receives the POST and calls the appropriate Receiving script or program to:

- parse the incoming HTTP Request,
- create the Receipt timestamp using the current date and time, and
- create an HTML Response to the Client.

An Internet ET Receiving Program may be implemented using a variety of technologies and techniques, including Active Server Pages (ASP), Common Gateway Interface (CGI), Java Server Pages (JSP), Java Servlets, and Personal Home Pages (PHP). The Internet ET is supported by most commercially available Web/HTTP Servers.

The Receiving Program and Process

The Receiving Program must be able to parse the multi-part form. It accomplishes this by finding the boundary string in the 'content-type' header and scanning for its occurrences further within the uploaded stream. Upon finding these boundary strings, the program must next determine the 'content-disposition' for each data element. This allows detection of the required text elements as well as the Internet ET payload file.

The Receiving Program only stores the payload file and is not concerned with the content of the payload file, which is encrypted. It will use the 'content-length' to determine how much data to expect in the body of the package.

A Receiving process requires an executable program or module that is called by the Server when it is identified by a POST operation.

When the Server receives a POST it will first read the header and populate environment variables before calling the Receiving Program. Most HTTP Servers read header variables and populate environment variables. Check the installed HTTP Server documentation for more information.

EXAMPLE: A SAMPLE HTTP POST HEADER

```
POST /cgi-bin/AS2dispatcher HTTP/1.1
Referer: http://www.get.a.life/upl.htm
Connection: Keep-Alive
User-Agent: brow v0.1 XYZ Corp.
Host: localhost
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, */*
content-type: multipart/form-data; boundary=-----87453838942833
Content-Length: 5379
```

After reading the HTTP header information, the Server will buffer the remaining data transmitted and call the Receiving Program specified in the POST statement. Do not assume that the Receiving Program is called as soon as the header is read, which can impact a receipt timestamp. The more common implementations buffer the entire transmission before calling the program. Check installed server implementation documentation.

The Receiving Program will have the following data stream available, and will have most of the header data available in environment variables.

EXAMPLE: DATA STREAM AVAILABLE TO RECEIVING PROGRAM

```
-----87453838942833
content-disposition: form-data; name="from"
123456789
-----87453838942833
content-disposition: form-data; name="to"
234567890
-----87453838942833
content-disposition: form-data; name="version"
1.9
-----87453838942833
content-disposition: form-data; name="receipt-disposition-to"
123456789
-----87453838942833
content-disposition: form-data; name="receipt-report-type"
gisb-acknowledgement-receipt
-----87453838942833
content-disposition: form-data; name="input-format"
X12
-----87453838942833
content-disposition: form-data; name="input-data"; filename="c:\temp\smallnom.bin"
content-type: multipart/encrypted; boundary=--boundary2--200309090001;
protocol="application/pgp-encrypted"
----boundary2--200309090001
content-type: application/pgp-encrypted
Version: 1
----boundary2--200309090001
content-type: application/octet-stream
-----BEGIN PGP MESSAGE-----
Version: PGP 9.0

hQCMaZRG1pEOIOvdAQp+JMr0m/9+8yOL60Z9Vr6fFV81FCExB/o0xmwiMkiwYsHsz0e8sb7ErC340MrNA/dw3taGM
jmI+CXyRF/PLEdg1NZE1ZCtNeL4YdIHAMLWwODG1QxhSucz8rMSgQ5mZzcOJwBdWLW70efgsu/9U1juJjYcluz6C0
3eFQv/43fkB+alAttgYdxX4g8QK664ad+Jo/XUICSmWBL66fqJR1KLeLf4wTaqGy174Aq48Wpvg1Eh785zC03UAW
0qg0ugMt86dPeyd91e2JigqwDYef/DYEKD0J9BGiGpS/uAupNKj8Ocp2IWClxKOGUbxpVNOntqWHS/GntegvDE/7
/ewCxDxsnmQS95p01141QZ1RQbeN.aqx2Dq/ra9g65HNchOCzjul5Vi8HHf6Yhg2WnROe+npByyCue6rihqgNVOJw
j0cVzpb4JE+gMDf3q4ISUb1Fv7/+SSFHDdnhdC5Ytpqf1Bc3B07hiLmtTXqNit31EbX9UVElObzSa9ZhxbC6/eS17
Nuf5ZTDsh9nrk+QQJ6FeC9W4cqXLj7IZySaR08Vtff+4ktqeuYusT4kSpnk027aw40/5jomUkfb22CAe4=
=Oiuo
-----END PGP MESSAGE-----
----boundary2--200309090001--
-----87453838942833--
```

This Receiving Program should check for basic validity in the environment variables and the data stream, and then parse the variables/data from the format. Data validations should include:

- The 'REQUEST_METHOD' environment variable is 'POST'.
- The 'CONTENT_TYPE' environment variable should be 'multipart/form-data' and the boundary, which cannot appear anywhere in the transaction being sent.
- The input stream should support binary mode to accommodate encrypted files.
- Each data element should be preceded by the boundary with the required two hyphen characters appearing before it.
- Each data element should contain the correct name on the 'content-disposition' line.
- Each data element should have a blank line ('\r\n\r\n' in C+ notation) before the start of the data.
- All tag values in the HTTP header should be evaluated in a case insensitive manner.
- Improperly formatted input. Finding the end of the stream using both 'content-length' and the boundary string terminator end mark is a good method to detect improperly formatted input.

Acknowledgement Receipt: 'gisb-acknowledgement-receipt'

The Acknowledgement Receipt ('Receipt') is critical to non-repudiation and business process timing. Immediately after the Receiving Program receives the last byte of data from the Sender, the Receiving Program should record the time and construct a 'gisb-acknowledgement-receipt'. This Receipt is sent from the Receiving Program to the Client prior to closing the HTTP connection.

The Receipt is a MIME-formatted text stream that includes the HTTP Response data elements (time-c, request-status, server-id, trans-id) in a 'multipart/report' MIME envelope.

The 'gisb-acknowledgement-receipt' including the 'multipart/report' envelope, is digitally signed, producing an 'application/pgp-signature' body part. As shown in the example below, the gisb-acknowledgement-receipt content, including the content boundaries, (lines 08 through 25) is signed to create the PGP payload (lines 29 to 37). Both the 'multipart/report' 'gisb-acknowledgement-receipt' and the 'application/pgp-signature' body parts are placed in a 'multipart/signed' envelope and the entire package is returned to the Sender.

EXAMPLE: SIGNED CONTENT FOR RESPONSE- LINE NUMBERS ADDED FOR ILLUSTRATION ONLY

```
01 content-type:multipart/signed; micalg=pgp-md5; protocol="application/pgp-signature";
02 boundary=--boundary2--200309090001
03
04 ----boundary2--200309090001
05 content-type: multipart/report; report-type="gisb-acknowledgement-receipt";
06 boundary="NAESB7867"
07
08 --NAESB7867
09 content-type: text/html
10
```

```
11 <HTML><HEAD><TITLE>Acknowledgement Receipt Success</TITLE></HEAD> <BODY><P>
12
13 time-c=19960619082855*
14 request-status=ok*
15 server-id=coolhost*
16 trans-id=234423897*
```

```
17 </P> </BODY></HTML>
18
19 --NAESB7867
```

```

20 content-type: text/plain.
21 time-c=19960619082855*
22 request-status=ok*
23 server-id=coolhost*
24 trans-id=234423897*
25 --NAESB7867--
26 ----boundary2--200309090001
27 content-type: application/pgp-signature
28
29 -----BEGIN PGP MESSAGE-----
30 Version: 2.6.26.5
31
32iQCVAwUBMJrRF2N9oWBghPDJAE9UQQAt17LuRVndBjrk4EqYBIb3h5QXIX/LC//JV5bNvkZIGPICEmI5iFd9boE
33gvpirHtIREEqLQRkYNoBActFBZmh9GC3C041WGquMbrbxc+nIs1TIK1A08rVi9ig/2Yh7LFrK5Ein57U/W72vgS3
34Lhe/zhdfolT9BrnHOxEa44b+EI=
35=ndaj
36
37 -----END PGP MESSAGE-----
38 ----boundary2--200309090001--

```

The Receipt name 'gisb-acknowledgement-receipt' retains the 'gisb-' prefix to assure compatibility with legacy GISB EDM implementations. The name is only used in the 'report-type' data element for the MIME part.

Additional Receiving Program Functions

- All data element names of the HTTP Request and Response fields will be in lower case. Note that the Internet ET standard format file contained in the Request and Response may follow a different standard.
- Carriage returns and line feeds will be ignored in all files.
- A field delimiter of '*' will be used in the HTTP Response. Please refrain from displaying a '*' anywhere else in the response so as not to confuse programs that need to parse on this basis.
- No spaces should surround the equal sign or the field delimiter.
- The required data elements must appear first in the HTTP Response and in the order specified. Additional information can be included after the required elements at the server's discretion.
- The first occurrence of the field name within the response will contain the value.
- If an HTML response is given, all data must be presented in a user-readable fashion. For example, if the required machine-readable fields are embedded in comments, another representation of these fields must be presented to the user.

Receiving Process URL Implementation Guidelines

Each company must offer at least one URL to accept files using Internet ET. Companies can offer multiple URLs. Though companies are free to construct a Web site with multiple 'single-purpose' URLs (e.g. nominations.xyzcorp.com; enrollments.xyzcorp.com) NAESB recommends the use of one 'general-purpose' URL.

The Receiving Program may initiate Error Notifications after the 'gisb-acknowledgement-receipt' is sent (e.g. file decryption errors). Error Notifications posted to the Sender would be directed to the Sender's general-purpose URL.

All URLs that will be required for use in the Internet ET process must be agreed to and defined in the TPW.

HTTP Response 'gisb-acknowledgement-receipt' Data Elements

Required HTTP Response Data Elements (listed in the required order)
--

time-c request-status server-id trans-id

Examples of HTTP Response Required Data Elements:

EXAMPLE: RESPONSE, SUCCESSFUL, MULTIPART FORMAT:

```
content-type: multipart/report; report-type="gisb-acknowledgement-receipt";  
boundary="NAESB7867"  
  
--NAESB7867  
content-type: text/html  
  
<HTML><HEAD><TITLE>Acknowledgement Receipt Success</TITLE></HEAD> <BODY><P>  
time-c=19960619082855*  
request-status=ok*  
server-id=coolhost*  
trans-id=234423897*  
</P> </BODY></HTML>  
--NAESB7867  
content-type: text/plain  
  
time-c=19960619082855*  
request-status=ok*  
server-id=coolhost*  
trans-id=234423897*  
--NAESB7867--
```

EXAMPLE: RESPONSE, ERROR, MULTIPART FORMAT:

```
content-type: multipart/report; report-type="gisb-acknowledgement-receipt";
boundary="NAESB7866"

--NAESB7866
content-type: text/html

<HTML><HEAD><TITLE>Acknowledgement Receipt Error</TITLE></HEAD> <BODY><P>
time-c=19960619082855*
request-status=EEDM106: Invalid To Common Code Identifier*
server-id=coolhost*
trans-id=234423897*
</P> </BODY></HTML>
--NAESB7866
content-type: text/plain

time-c=19960619082855*
request-status=EEDM106: Invalid To Common Code Identifier*
server-id=coolhost*
trans-id=234423897*
--NAESB7866--
```

EXAMPLE: RESPONSE, WARNING, MULTIPART FORMAT:

```
content-type: multipart/report; report-type="gisb-acknowledgement-receipt";
boundary="NAESB7866"

--NAESB7866
content-type: text/html

<HTML><HEAD><TITLE>Acknowledgement Receipt Warning</TITLE></HEAD> <BODY><P>
time-c=19960619082855*
request-status=WEDM100: Transaction Set Sent, Not Mutually Agreed*
server-id=coolhost*
trans-id=234423897*
</P> </BODY></HTML>
--NAESB7866
content-type: text/plain

time-c=19960619082855*
request-status= WEDM100: Transaction Set Sent, Not Mutually Agreed *
server-id=coolhost*
trans-id=234423897*
--NAESB7866--
```

EXAMPLE: RESPONSE, SUCCESSFUL, SIGNED RECEIPT:

```
content-type:multipart/signed; micalg=pgp-md5; protocol="application/pgp-signature";
boundary=--boundary2--200309090001

----boundary2--200309090001
content-type: multipart/report; report-type="gisb-acknowledgement-receipt";
boundary="NAESB7867"

--NAESB7867
content-type: text/html

<HTML><HEAD><TITLE>Acknowledgement Receipt Success</TITLE></HEAD> <BODY><P>

time-c=19960619082855*
request-status=ok*
server-id=coolhost*
trans-id=234423897*

</P> </BODY></HTML>

--NAESB7867
content-type: text/plain.
time-c=19960619082855*
request-status=ok*
server-id=coolhost*
trans-id=234423897*
--NAESB7867--
----boundary2--200309090001
content-type: application/pgp-signature

-----BEGIN PGP MESSAGE-----
Version: 2.6.26.5

iQCVAwUBMJrRF2N9oWBghPDJAE9UQQAt17LuRvndBjrk4EqYBIb3h5QXIX/LC//JV5bNvkZIGPICEmI5iFd9boEg
vpIrhTIREEqLQRkYNoBActFBZmh9GC3C041WGquMbrbxc+nIs1TIKlA08rVi9ig/2Yh7LFrK5Ein57U/W72vgSxLh
e/zhdfo1T9BrnHOxEa44b+EI=
=ndaj

-----END PGP MESSAGE-----
----boundary2--200309090001--
```

EXAMPLE: HTML FORMAT RESPONSE, SUCCESSFUL:

```
<html><head><title>Upload OK</title></head>
<body>
<!-- time-c=19960123203618*-->
<!-- request-status=ok* -->
<!-- server-id=coolhost*-->
<!-- trans-id=232323897*-->
<h1>Upload OK</h1>
<b>File Saved at (time-c):</b> 19960123203618<br>
<b>Status (request-status):</b>ok<br>
<b>Server (server-id):</b>coolhost<br>
<b>Transaction ID (trans-id):</b>232323897<br>
</body></html>
```

SENDING INTERNET ET ERROR NOTIFICATION

When a Client sends an Internet ET package to a Server, the Server responds with a Receipt. Further back-office processing (e.g. decryption) may be required, and additional errors may be found.

Error Notification transactions are used to communicate transport errors found by the Receiver after the initial receipt is sent to the Sender.

Errors from translation and other back-office processing are outside the scope of the Internet ET.

When a file passes the decryption step, no Error Notification is sent back to the Client. If the decryption step fails, an Error Notification must be sent to the Client.

The Error Notification format applies to the posting of an error message after the Sender's session has been disconnected. This Error Notification is used only if the original HTTP Response is returned with an 'ok'.

Required Error Notification Data Elements

The data elements for the Error Notification are the same as those described in the Section 'Sending Internet ET Packages', with the exception of the 'input-format' and 'input-data' elements. The file containing the data elements for Error Notification should not be encrypted.

Required Data Elements for Error Notification (listed in the required order):

1. 'from'
2. 'to'
3. 'input-format'

Error Notification 'input-data' Element Specifications:

- The file containing the data elements for Error Notification should not be encrypted.
- All data element names will be in lower case in the Error Notification.
- Carriage returns and line feeds will be ignored in all files.
- A field delimiter of '*' will be used in the Error Notification. Please refrain from displaying a '*' anywhere else in the Error Notification so as not to confuse programs that need to parse on this basis.
- No spaces should surround the equal sign or the field delimiter.
- The required data elements must appear first in the response.
- Additional information can be included after the required elements at the server's discretion.
- The first occurrence of the field name within the response will contain the value.
- An Error Notification contains two body parts nested within a multipart/report outer envelope with the content-type of 'gisb-error-notification'.
- The first body part contains human readable content in HTML. The second body part contains machine readable content in plain text.
- The multipart/report containing the Error Notification is used to create a digital signature body part, identified by a 'content-type' of application/pgp-signature. Both the multipart/report Error Notification and application/pgp-encrypted digital signature body parts are combined in a multipart/signed envelope.

EXAMPLE: ERROR NOTIFICATION INTERNET ET PACKAGE:

```

POST /cgi-bin/AS2dispatcher HTTP/1.1
Referer: http://www.acmeenergy/upl.htm
Connection: Keep-Alive
User-Agent: brow v0.1 XYZ Corp.
Host: localhost
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, */*
content-type: multipart/form-data; boundary=-----87453838942833
Content-Length: 1958

-----87453838942833
content-disposition: form-data; name="from"

234567890
-----87453838942833
content-disposition: form-data; name="to"

123456789
-----87453838942833
content-disposition: form-data; name="version"

1.9
-----87453838942833
content-disposition: form-data; name="receipt-disposition-to"

123456789
-----87453838942833
content-disposition: form-data; name="receipt-report-type"

gisb-acknowledgement-receipt
-----87453838942833
content-disposition: form-data; name="input-format"

error
-----87453838942833
content-disposition: form-data; name="input-data"; filename="c:\temp\error.not"
content-type: multipart/signed; micalg=pgp-md5; protocol="application/pgp-signature";
boundary=--boundary2--200309090001

----boundary2--200309090001

content-type: multipart/report; report-type="gisb-error-notification";
boundary="NAESB7868"

--NAESB7868
content-type: text/html

<HTML><HEAD><TITLE>Error Notification</TITLE></HEAD> <BODY><P>
orig-from=123456789*
orig-to=234567890*
orig-input-format=X12*
resp-time-c=19960619102855*
resp-server-id=coolhost*
resp-trans-id=234423897*
request-status=EEDM601: Public Key Invalid*
comments=Please contact 1-800-555-1212 for correct public key*

</P> </BODY></HTML>

--NAESB7868
content-type: text/plain

orig-from=123456789*
orig-to=234567890*
orig-input-format=X12*
resp-time-c=19960619102855*
resp-server-id=coolhost*
resp-trans-id=234423897*
request-status=EEDM601: Public Key Invalid*
comments=Please contact 1-800-555-1212 for correct public key*

```

```
--NAESB7868--
-----boundary2--200309090001

content-type: application/pgp-signature

-----BEGIN PGP MESSAGE-----

Version: 6.5

iQCVAwUBMJrRF2N9oWBghPDJAE9UQQAt17LuRVndBjrk4EqYBIb3h5QXIX/LC//JV5bNvkZIGPIcEmI5iFd9boEg
vpirHtIREEqLQRkYNoBActFBZmh9GC3C041WGQuMbrbxc+nIs1TIK1A08rVi9ig/2Yh7LFrK5Ein57U/W72vgSxLh
e/zhdfo1T9BrnHOxEa44b+EI=
=ndaj
-----END PGP MESSAGE-----
-----boundary2--200309090001--
```

Pre-validation before Decryption

Proper trapping of the range of decryption process errors listed in Table 1 (Internet ET Standard Error Codes and Messages) may require program code which is external to the decryption algorithm. Some versions of the PGP software do not explicitly discriminate between EEDM601, EEDM602, EEDM603, and EEDM699 type errors.

Under such a circumstance, files inbound to the decryption process should be preprocessed to trap the errors not identified by the PGP/OpenPGP version being used. For example, searching the file for the text strings 'BEGIN PGP MESSAGE' and 'END PGP MESSAGE' can quickly identify 'EEDM602 File not encrypted' and 'EEDM603 Encrypted file truncated' type errors when the implemented PGP version only identifies decryption success, invalid Public Key (EEDM601), and decryption failure (EEDM699).

SECURITY

Internet ET security requirements include four primary security aspects: data Privacy, data Integrity, Authentication, and Non-repudiation (PAIN).

- Data privacy: unauthorized parties cannot decipher the content of the data.
- Authentication: the Receiver is certain of the identity of the Sender.
- Data integrity: unauthorized parties cannot modify or corrupt the data.
- Non-repudiation: the Sender cannot deny ownership of the transaction if it was sent with their digital signature.

In general, these needs are met by using the HTTP Basic Authentication using Transport Layer Security capability of the Web server, and the encryption and digital signature capability of the OpenPGP and PGP security application for securing transactions. Additional issues such as firewall security are discussed in the standards, and specific security implementation details should be addressed by each organization.

Though many decisions as to overall security measures are left to each Trading Partner and their environment, security standards have been created to ensure a minimum level of confidence in conducting business over the Internet and to provide some uniformity in the implementation of security.

Understanding OpenPGP and PGP

Pretty Good Privacy (PGP) is the name of a proprietary encryption software chosen for Internet ET to both digitally sign and encrypt payloads being sent. OpenPGP is the Internet Engineering Task Force (IETF) standard version of PGP that excludes all patented algorithms, allowing free commercial use of the standard. Both OpenPGP and PGP use a Public Key/Private Key pair to secure and sign files for transfer. The Private Key must be known only to the company that generated it. The Public Key counterpart is shared with Trading Partners.

Each company must generate its Public Key and Private Key pair. The RSA key generation algorithm should be chosen for versions of PGP that offer alternatives. Implementers of OpenPGP should choose DSA and El Gamal when creating their key pair. The Public Keys should be distributed electronically to the company's Trading Partners in a secure manner.

Never divulge a Private Key to another party. If an untrusted party has your Private Key, your security is compromised.

When a company wishes to send transactions to its Trading Partner, it will use the partner's Public Key to encrypt the file. Encryption provides data privacy. Only the Private Key counterpart can decrypt this file.

When the sending party encrypts the file, it also uses its own Private Key to 'sign' the transaction. The Receiving Party can use the Sender's Public Key to verify the signature. The digital signature provides non-repudiation.

Encryption / Digital Signature

Encryption and digital signatures are applied to Payload Files before they are sent by the Batch Browser. The use of internal file or payload encryption (including but not limited to X12.58 encryption) is outside the scope of NAESB encryption standards but does not conflict with OpenPGP/PGP.

Encryption and digital signatures are created using OpenPGP, or on a mutually agreed basis, PGP (see Appendix A for minimum version). Regardless of encrypting in a manual or automated fashion, it is essential that the correct Public Key of the Trading Partner be used to encrypt and just as essential that the correct Sender's own Private Key be used to digitally sign the file.

Decryption / Digital Signature Verification

After a package is received and processed by the Receiving Program, it is ready to be decrypted and have its digital signature verified. Given the correct userID for a Trading Partner, OpenPGP/PGP uses the appropriate key pair to encrypt, sign and decrypt. Upon request for signature verification, the OpenPGP/PGP will return a human-readable descriptive text such as DUNS number or company name.

The digitally signed HTTP Response received by the Sender of the transaction should be verified to ensure non-repudiation of receipt of the transaction.

Throughput Considerations

Encryption, digital signing, decryption and signature verification are all very CPU intensive. Companies anticipating large volumes of Internet ET traffic should research state-of-the-art techniques for scalability, including but not limited to:

- separating decryption and signature verification processing from web server receiving and processing,
- passing secured or to-be-secured packages to a separate computer for security processing,
- optimizing CPU and memory on security processing computers, and
- real-time or near real-time monitoring of website performance.

Implementers of Internet ET sites should review and evaluate Domain Name Server (DNS) cache refresh intervals to ensure Trading Partner address changes are recognized on a timely basis.

Decryption and digital signature verification may not necessarily be processed by the Receiving Program prior to the 'gisb-acknowledgement-receipt' being sent to the Sender. As a result, the Sender may get an HTTP Response indicating a successful transfer but still not know if the file was successfully decrypted by the Receiver. Guidelines for communicating decryption errors found after the initial HTTP Response is sent are in Section 'Sending Error Notification Transactions' and Table 1, 'Internet ET Standard Error Codes and Messages'.

Security Requirements

Basic Authentication. Basic authentication, also known as realm one security, has been defined as one of the security standards for transmission on the Internet. The userid and password will be assigned by the server party according to site standards. The TPA must identify the userid and password for this security as well as procedures for changing the password, if applicable.

OpenPGP or PGP File Encryption. Payload Files are encrypted using OpenPGP, or on a mutually agreed basis, PGP (see Appendix A for minimum version).

Firewall. A firewall should be deployed to protect HTTP Servers.

CLIENT AND SERVER SPECIFICATIONS

Synchronization. Each Client and Server should be synchronized to a clock in the network of atomic clocks that is accessible via the Internet. The Client and Server should be synchronized as necessary to ensure synchronization with an atomic clock +/- 5 seconds. Please refer to Appendix A, 'Time Synchronization' for references on public sites for synchronization.



TESTING GUIDELINES

NAESB INTERNET ET TEST GUIDELINES

Implementation of Internet ET requires testing to assure all parties are prepared to operate according to the Internet ET. This document focuses on testing standards for establishing Internet ET connectivity with a Trading Partner. Testing for transaction and other relevant business processes can be found in the WGQ QEDM.

Internet ET Connectivity testing standards may include:

- Connectivity test scripts. These scripts define the steps needed to adequately test connectivity.
- TPW is a worksheet defines important operations parameters for a Trading Partner including testing parameters. The parameters include Internet ET URLs, contacts, and other information.

Common Internet ET errors include:

- Misspelled keywords (e.g. 'content-type'), or spacing in a keyword,
- Header 'content-type' missing,
- MIME boundary not correct,
- Malformed MIME segments,
- Content-length does not match actual length, and
- PGP MIME malformed (found with some versions of PGP/OpenPGP).

GENERAL TESTING ASSUMPTIONS

The following assumptions apply to Internet ET Testing:

- This document covers Internet ET Testing. Transactions and business process test plans can be found in the WGQ QEDM.
- Testing may uncover problems. Problems found during testing should be expected.
- Testing is a basic demonstration of competency, and may not uncover all problems that may eventually require correction.
- In normal circumstances, Trading Partner to Trading Partner Internet ET connectivity testing takes approximately two weeks.

TESTING GOALS

The primary testing goals of this Internet ET are:

- Establish Internet ET connectivity between Trading Partners including Internet connections and encryption compatibility.
- Validate Internet ET header formatting and delimiters.
- Validate that normal production transaction files can be delivered.
- Validate that a large file (1MB or larger) can be delivered.
- Validate that Internet ET Receipts ('gisb-acknowledgement-receipt') are being delivered.
- Validate that Protocol Failures are handled properly.
- Validate that Exchange Failures are handled properly.
- Validate that encryption/decryption and digital signature failures are handled properly.

TEST EXECUTION

Test Scripts

Test scripts provide a step-by-step process for testing Trading Partner Internet ET connectivity. Test script scenarios test for both positive (accept) and negative (reject) results. Typical test scripts involve an exchange (Request and Response) of data between Trading Partners, with each TP confirming receipt of test file exchange via normal Internet ET standards. A copy of the payload file can be sent via e-mail for verification.

Test scripts can validate:

- That received files were not corrupted.
- Fail-over mechanisms by simulating a Protocol Failure and an Exchange Failure, triggering the appropriate notices to the TP contacts.
- Encryption failure processes by simulating an encryption/decryption failure, triggering the appropriate notices to the TP contacts.
- System time clock synchronization.

Recommended Internal Tests

In addition to tests executed with Trading Partners, the following tests are recommended as internal tests of Internet ET systems.

- Acquire or develop an HTML page for interactive file upload (sample code is earlier in this document). Test the interactive file upload using an Interactive Browser to determine success.
- Stress Test. Ability to send and receive large production files (e.g. 10MB minimum uncompressed) and simultaneous usage. Simultaneous loading can be tested by requesting several Trading Partners and/or internal parties to simulate concurrent Internet ET transfers.
- Fail-over Test. Test any processes triggered by a Protocol Failure or Exchange Failure with Trading Partner.
- Invalid Userid/Passwords. Thoroughly test using the incorrect userid and password against the secure directory.
- Simulated Errors. Test various simulated errors in both file transfers and in OpenPGP or PGP decryption.

APPENDICES

TABLE 1 – INTERNET ET STANDARD ERROR CODES AND MESSAGES

These errors and warnings are strictly related to problems found in the Receiving Program or decryption levels of processing before translation. Errors and warnings generated by the Client Batch Browser are assumed to be documented at the Client site to distinguish them from problems occurring in the Receiving Program or decryption. Numbering schemes and descriptions should aid in this distinction.

EEDM###	standard error format with ### representing a numeric value; further processing will not take place
WEDM###	standard warning format with ### representing a numeric value; further processing will take place

The string for the error or warning should appear in the following format:

[Validation Code]:[Description];[supplemental message to be defined by the issuing site up to 80 characters] ®

Internet ET Standard Error Codes and Messages

Validation Code	Description	Data Element	Data Element Required or Mutually Agreed
EEDM100	Missing 'from' Common Code Identifier code	from	required
EEDM101	Missing 'to' Common Code Identifier	to	required
EEDM102	Missing input format	input-format	required
EEDM103	Missing data file	input-data	required
EEDM104	Missing transaction set	transaction-set	mutually agreed
EEDM105	Invalid 'from' Common Code Identifier	from	required
EEDM106	Invalid 'to' Common Code Identifier	to	required
EEDM107	Invalid input format	input-format	required
EEDM108	Invalid transaction set	transaction-set	mutually agreed
EEDM109	No parameters supplied	parameter string	required
EEDM110	Invalid 'version'	version	required
EEDM111	Missing 'version'	version	required
EEDM113	Invalid 'receipt-security-selection'	receipt-security-selection	required
EEDM114	Missing 'receipt-disposition-to'	receipt-disposition-to	required
EEDM115	Invalid 'receipt-disposition-to'	receipt-disposition-to	required
EEDM116	Missing 'receipt-report-type'	receipt-report-type	required
EEDM117	Invalid 'receipt-report-type'	receipt-report-type	required
EEDM118	Missing 'receipt-security-selection'	receipt-security-selection	required
EEDM119	Mutually agreed element, refnum, not present	refnum	mutually agreed
EEDM120	Mutually agreed element refnum-orig not present	refnum-orig	mutually agreed
EEDM121	Duplicate refnum received	refnum	mutually agreed
EEDM601	Public key invalid	file itself	required - security
EEDM602	File not encrypted	file itself	required - security
EEDM603	Encrypted file truncated	file itself	required - security
EEDM604	Encrypted file not signed or signature not matched	file itself	required - security

Validation Code	Description	Data Element	Data Element Required or Mutually Agreed
EEDM699	Decryption Error		required for general decryption errors not specifically identified by OpenPGP or PGP messages or exit codes
EEDM701	Sending party not associated with Receiving party		required
EEDM702	Package file format not recognized by Receiving party		required when the file format is not recognized by the receiver (e.g. not expecting 855 or not expecting Flat-File or XML)
EEDM703	Data set exchange not established for Trading Partner		required if the translator does not handle this exception
EEDM999	System error		required for general system errors to indicate severe errors in processing at the receiving site
WEDM100	Transaction set sent not mutually agreed	transaction-set	mutually agreed
WEDM103	Missing 'receipt-security-selection'	receipt-security-selection	required
WEDM104	Element refnum received, not mutually agreed; ignored	Refnum	mutually agreed
WEDM105	Refnum-orig received but not mutually agreed; ignored	refnum-orig	mutually agreed

APPENDIX A - Reference Guide

HTTP

The NAESB Internet ET architecture is based on HTTP 1.1, and all implementations should be compatible with this version. All aspects of HTTP, HTML, and other Web-related topics are documented at: <http://www.w3.org>.

General information regarding HTTP with basic terminology included are documented at: <http://tools.ietf.org/html/rfc7230>.

HTTP multipart form file upload (RFC 2854).

Syntax information for multipart can be found in IETF RFC7578.

HTML

Effective March 27, 2018, the World Wide Web Consortium has made a recommendation for HTML 4.01. Information on HTML 4.01 may be found at: <http://www.w3.org/TR/REC-html401/>

Payload Encryption and Digital Signatures

For document level security (digital signature and encryption), OpenPGP is the standard method for the Internet ET; however, PGP can be used on a mutually agreed basis. The minimum asymmetric key length for either OpenPGP or PGP is 2048 bits.

OpenPGP

Open PGP is an open source security solution and is available for a variety of operating systems and platforms. The IETF OpenPGP standard is preferred for Internet ET payload encryption and digital signatures. More information on the current version is available at <http://www.iefr.org/rfc/rfc4880.txt>.

PGP

Although OpenPGP is the preferred method for payload encryption and digital signatures, Symantec offers PGP (minimum version 9) as a comparable, commercial, off-the-shelf solution and can be used on a mutually agreed basis. If PGP version 9 is compromised, the latest Generally Available version should be implemented within nine months of such version being supported by major development platforms. More information is available at <https://www.symantec.com/products/encryption>.

Time Synchronization

Time synchronization is required to assure that all Trading Partners' transaction times are accurate. Time accuracy is dependent on how much a system's clock drifts, how frequently it is resynchronized and the accuracy of the source used for synchronization.

Each NAESB business process may have unique time-synchronization requirements. Servers need to be time-synchronized according to the standards needed for the most-restrictive target market (i.e. smallest drift allowance).

Authoritative time synchronization provided by governmental agencies around the world based on a synchronized network of atomic clocks. In the United States this includes the U. S. Naval Observatory and the National Institute of Standards and Technology.

One method of obtaining the current time is from the U. S. Naval Observatory's Web site (see below reference). The output from this page can easily be edited and reformatted to set a local system's time. Commercial, shareware and public domain packages are also available to synchronize system times.

Further information on time synchronization may be found at the following Web sites:

- <https://www.usno.navy.mil/USNO/time/display-clocks/simpletime>
- <https://www.time.gov/>
- <https://www.pool.ntp.org/en/>



APPENDIX B - Minimum Technical Characteristics and Guidelines for the Developer and User of the Customer Activities and Informational Postings Web Sites¹

The below listed minimum technical characteristics provide specifications to the developer of the user environment for which the application will be designed and tested. Likewise, they will serve as guidelines to the user when purchasing the appropriate hardware and software to enable him/her to use the application.²

Browser:

The latest Generally Available (GA) versions of both Google Chrome^{®3} and Microsoft Edge^{®4} within 9 months of such GA version becoming available, characteristics including:

Cookies

Frames & Nested Frames

HTML

JavaScript

Style Sheets

Tables & Nested Tables

Users may continue to browse with Microsoft Internet Explorer^{®4} and Mozilla FireFox^{®5} Extended Support Release for those Customer Activities Web Sites compatible with these Browsers.

Plug-ins (GA versions within 9 months of such GA versions becoming available)

Adobe Acrobat Reader^{®6} or equivalent PDF Reader

Independent Computer Architecture (ICA^{®7}) – Protocol used for remote control access to an application

Operating Systems:

Operating systems on a Client workstation should be multithreaded, pre-emptive (for example Windows^{®10}) and the latest, stable version and service pack available.

Hardware:

CPU	2 GHz
Connection	Broadband
Display Resolution	1280 x 1024, 16K colors
Memory	4 GB Physical

¹. Configuration shown indicates a minimum except where a specific level is established. 'Minimum' implies a level where a reasonable experience for the user may be achieved. These levels also indicate the level that a user may expect that a Client environment has been tested. Results may be less than satisfactory, or may preclude use of a site, if the user chooses to use anything less than those levels shown.

². This list in no way should be construed as an endorsement by NAESB WGQ of any specific products.

³. Google Chrome[®] is a registered trademark of Google, Inc.

⁴. Microsoft Edge[®] and Microsoft Internet Explorer[®] and Microsoft Windows[®] are registered trademarks of Microsoft Corporation.

⁵. Mozilla Firefox[®] is a registered trademark of Mozilla.

⁶. Adobe[®] Acrobat[®] and Reader[®] are registered trademarks of Adobe Systems Incorporated.

⁷. ICA[®] is a registered trademark of Citrix Systems Inc.

Security:

The latest GA version of Transport Layer Security (TLS), using a minimum 128-bit RSA Encryption, within 9 months of such GA version being supported by major development platforms.

TLS (RFC 7919 and RFC 8446)

HTTP Basic Authentication (RFC 7617)

Browsers – Users may continue to browse with Microsoft Internet Explorer^{®3} and FireFox^{®2} Extended Support Release for those Customer Activities Web Sites compatible with these Browsers.

Informational Postings Web Site Developer Special Technical Considerations

User's environment supporting the above minimum characteristics should be able to access all NAESB WGQ standardized features of Informational Postings Web Sites. Any other Web technologies may be considered for use by the Informational Postings developer as long as they can be used by the Client without requiring special actions including, but not limited to, the following:

- Requiring firewall rule changes that are not specified in Appendix C,
- Requiring use of a specific Browser not specified in this Appendix,
- Requiring a UserID/Password, and
- Requiring download of special helper applications (plug-ins, viewers or readers, etc.) not specified in this Appendix.

Appendix C - Minimum Technical Characteristics for EDM Communications

For securing EDM communications, Trading Partners should utilize HTTP Basic Authentication (RFC 7617) and the latest Generally Available version of Transport Layer Security (TLS) (minimum 128-bit RSA encryption) within 9 months of such version being supported by major development platforms.

The following ports may be used by EDM developers and should be made available in user environments.

Allowable TCP Ports (not UDP ports):

HTTPS 443, 5713, 6112, 6304, 6874, 7403

ICA®¹ 1494

SMTP 25

Trading Partners may mutually agree to use ports other than those specified above. Only ports necessary to transaction-related traffic should be opened, and they should be protected by network and system security controls. All other ports should be closed. The Client-side firewall implementation and Client Browser settings should permit the downloading and installation of NAESB WGQ approved plug-ins and modules. Please refer to Appendix B of this manual the NAESB WGQ defined Minimum Technical Characteristics for Accessing Customer Activities Web Sites, for the listing of NAESB WGQ approved plug-ins and modules.

APPENDIX D – FREQUENTLY ASKED QUESTIONS

	Page
Q1: How many Internet ET package exchange attempts should occur before notifying a Trading Partner?	54
Q2: Can the 'gisb-acknowledgement-receipt' be sent before or after Internet ET package decryption?	55
Q3: What cryptographic algorithms should be used or not used?	55
Q4: NAESB EDM / AS2 Compatibility. What is the status of NAESB compatibility with AS2?	55
Q5: Atomic Clock Synchronization. How often should Internet ET system clocks synchronize with an atomic clock?	55
Q6: Internet Continuous Connection. Does an end user need a continuously-connected internet Web server to participate in the Internet ET , or can a dial-up connection using supported Browser software be sufficient?	55
Q7: Use of ANSI X12.58. If ANSI X12.58 encryption is used, is OpenPGP or PGP encryption still required?	55
Q8: What does NAESB recommend for the Open PGP/PGP descriptive text?	55
Q9: What does NAESB say about an industry participant's security?	56
Q10: Does NAESB require whitelisting?	56

Q1: How many Internet ET package exchange attempts should occur before notifying a Trading Partner?

A: The Internet ET 'Exchange Failure' standard requires at least three package exchange attempts over a 30- to 120-minute period. At minimum, this means 30 minutes has elapsed between the first failed attempt and the third failed attempt. At maximum, 120 minutes can elapse between the first failed attempt and the third failed attempt. Notifying your Trading Partner must occur if the time between the first failed attempt and the last failed attempt is greater than 120 minutes.

For example, the first attempt occurs at 00:30:00, and the third attempt at time 00:30:00; the second attempt can occur any time between the first and third. If the third attempt fails, there is an 'Exchange Failure' and the Trading Partner should be notified.

Q2: Can the 'gisb-acknowledgement-receipt' be sent before or after Internet ET package decryption?

A: Either. If an implementation decrypts packages after the 'gisb-acknowledgement-receipt' is sent, errors found must be communicated to Trading Partners using the Error Notification transaction. The TPW identifies when packages are decrypted by each Trading Partner.

Regardless of when decrypting occurs, the 'time-c' timestamp does not change. It is always the time the last byte was received by the Server from the Sender.

Q3: What cryptographic algorithms should be used or not used?

A: OpenPGP implementations should use DSA and El Gamal, and PGP implementations should use RSA.

Q4: NAESB EDM / AS2 Compatibility. What is the status of NAESB compatibility with AS2?

A: AS2 and NAESB EDM are no longer compatible. The GISB/NAESB EDM and AS2 standards were separated as of version 12 of AS2. The AS2 standard now supports the UCC profile, and not the GISB profile. At this time NAESB is not pursuing an IETF® standard for the Internet ET.

Q5: Atomic Clock Synchronization. How often should Internet ET system clocks synchronize with an atomic clock?

A: Systems should be synchronized as often as necessary to maintain the required +/- 5 second variance with the NIST atomic clock. Some business processes may require more stringent synchronization. Refer to quadrant standards for time-synchronization standards of business processes.

Q6: Internet Continuous Connection. Does an end user need a continuously-connected internet Web server to participate in the Internet ET or can a dial-up connection using supported Browser software be sufficient?

A: An intermittent connection using a supported Interactive Browser is not enough to actively participate in the Internet ET. Internet ET requires a Server with a permanent Internet connection capable of receiving files without operator intervention; this Server may exist at a service provider.

Q7: Use of ANSI X12.58. If ANSI X12.58 encryption is used, is OpenPGP or PGP encryption still required?

A: Yes. The use of encryption such as X12.58 on Payload Files is outside the scope of the NAESB encryption standards.

Q8: What does NAESB recommend for the OpenPGP/PGP descriptive text?

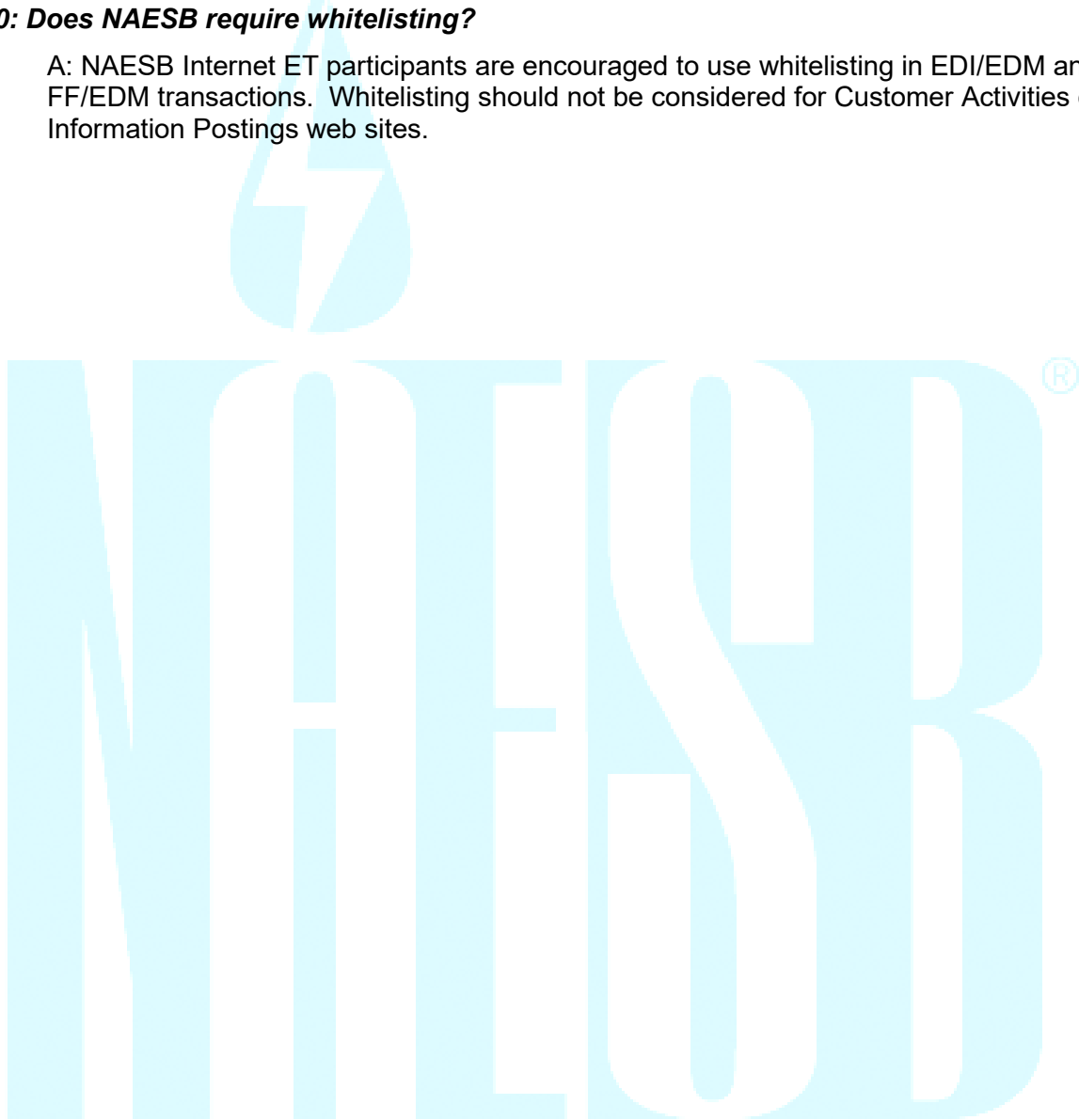
A: There are no Internet ET standards for the information provided in the OpenPGP/PGP descriptive text data element. Implementers are encouraged to use their company name in this data element.

Q9: What does NAESB say about an industry participant's security?

A: NAESB Internet ET participants are encouraged to maintain their system security in such a manner that reduces the risk of unauthorized/malicious activity. However, NAESB does not dictate overall security requirements for individual companies. For further information on general security guidelines please reference the SANS (www.SANS.com) or NIST (www.NIST.com) websites.

Q10: Does NAESB require whitelisting?

A: NAESB Internet ET participants are encouraged to use whitelisting in EDI/EDM and FF/EDM transactions. Whitelisting should not be considered for Customer Activities or Information Postings web sites.



VERSION NOTES

Version 4.0

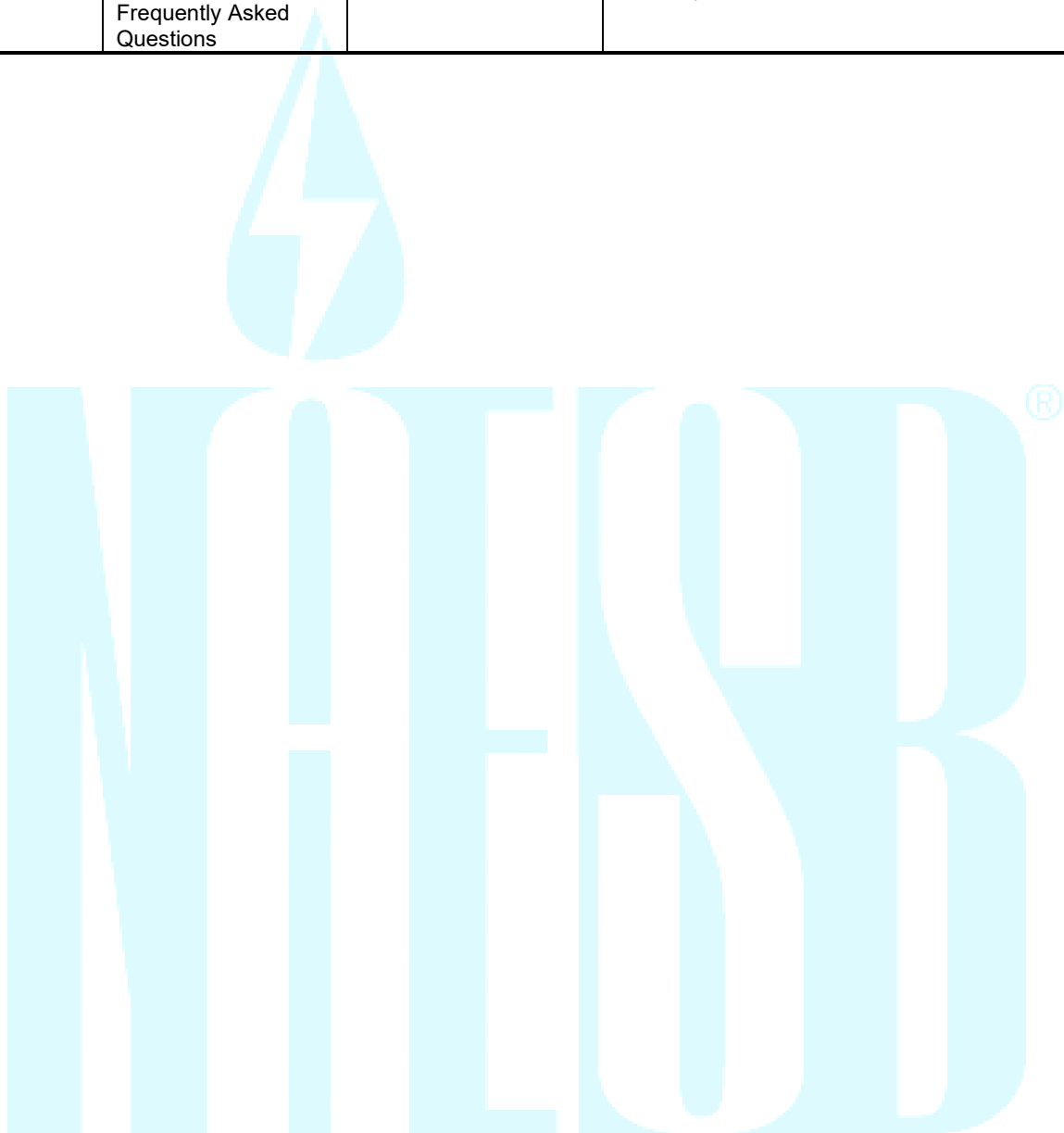
In MC22007, a new implementation guide was developed for security specific standards. The new Cybersecurity Related Standards implementation guide is a compilation of all security specific standards, and corresponding implementation information. This includes essentially the entire Version 3.2 of the WGQ/RMQ Internet Electronic Transport (Internet ET) manual, and all cybersecurity related standards and corresponding information from Version 3.2 of the WGQ QEDM manual. Additionally, this manual was developed to be WGQ specific. For more information on this effort, please see the Final Action for MC22007, as posted on the NAESB web site.

The following table shows a summary of modifications to Version 4.0 of the WGQ Cybersecurity Related Standards.

Standard	Description	Request No.	Action
	Executive Summary	MC22007	Modified to remove references to RMQ and Internet ET manual, and to add clarity to new purpose.
	Introduction	MC22007	Modified to remove references to RMQ and Internet ET manual, and to add clarity to new purpose.
12.1.1	Principles	MC22007	Formerly Internet ET Standard No. 10.1.1
12.1.2	Principles	MC22007	Formerly Internet ET Standard No. 10.1.2
12.1.3	Principles	MC22007	Formerly Internet ET Standard No. 10.1.3
12.1.4	Principles	MC22007	Formerly Internet ET Standard No. 10.1.4
12.1.5	Principles	MC22007	Formerly Internet ET Standard No. 10.1.5
12.1.6	Principles	MC22007	Formerly Internet ET Standard No. 10.1.6
12.1.7	Principles	MC22007	Formerly Internet ET Standard No. 10.1.8
12.1.8	Principles	MC22007	Formerly Internet ET Standard No. 10.1.10
12.1.9	Principles	MC22007	Formerly QEDM Standard No. 4.1.37
12.2.1	Definition	MC22007	Formerly Internet ET Standard No. 10.2.1
12.2.2	Definition	MC22007	Formerly Internet ET Standard No. 10.2.2
12.2.3	Definition	MC22007	Formerly Internet ET Standard No. 10.2.3
12.2.4	Definition	MC22007	Formerly Internet ET Standard No. 10.2.4
12.2.5	Definition	MC22007	Formerly Internet ET Standard No. 10.2.5
12.2.6	Definition	MC22007	Formerly Internet ET Standard No. 10.2.6
12.2.7	Definition	MC22007	Formerly Internet ET Standard No. 10.2.7
12.2.8	Definition	MC22007	Formerly Internet ET Standard No. 10.2.8
12.2.9	Definition	MC22007	Formerly Internet ET Standard No. 10.2.9
12.2.10	Definition	MC22007	Formerly Internet ET Standard No. 10.2.10
12.2.11	Definition	MC22007	Formerly Internet ET Standard No. 10.2.11
12.2.12	Definition	MC22007	Formerly Internet ET Standard No. 10.2.12
12.2.13	Definition	MC22007	Formerly Internet ET Standard No. 10.2.13
12.2.14	Definition	MC22007	Formerly Internet ET Standard No. 10.2.14
12.2.15	Definition	MC22007	Formerly Internet ET Standard No. 10.2.15
12.2.16	Definition	MC22007	Formerly Internet ET Standard No. 10.2.16
12.2.17	Definition	MC22007	Formerly Internet ET Standard No. 10.2.17
12.2.18	Definition	MC22007	Formerly Internet ET Standard No. 10.2.18
12.2.19	Definition	MC22007	Formerly Internet ET Standard No. 10.2.19
12.2.20	Definition	MC22007	Formerly Internet ET Standard No. 10.2.20
12.2.21	Definition	MC22007	Formerly Internet ET Standard No. 10.2.21
12.2.22	Definition	MC22007	Formerly Internet ET Standard No. 10.2.22
12.2.23	Definition	MC22007	Formerly Internet ET Standard No. 10.2.23
12.2.24	Definition	MC22007	Formerly Internet ET Standard No. 10.2.24

Standard	Description	Request No.	Action
12.2.25	Definition	MC22007	Formerly Internet ET Standard No. 10.2.25
12.2.26	Definition	MC22007	Formerly Internet ET Standard No. 10.2.26
12.2.27	Definition	MC22007	Formerly Internet ET Standard No. 10.2.27
12.2.28	Definition	MC22007	Formerly Internet ET Standard No. 10.2.28
12.2.29	Definition	MC22007	Formerly Internet ET Standard No. 10.2.29
12.2.30	Definition	MC22007	Formerly Internet ET Standard No. 10.2.30
12.2.31	Definition	MC22007	Formerly Internet ET Standard No. 10.2.31
12.2.32	Definition	MC22007	Formerly Internet ET Standard No. 10.2.32
12.2.33	Definition	MC22007	Formerly Internet ET Standard No. 10.2.33
12.2.34	Definition	MC22007	Formerly Internet ET Standard No. 10.2.34
12.2.35	Definition	MC22007	Formerly Internet ET Standard No. 10.2.35
12.2.36	Definition	MC22007	Formerly Internet ET Standard No. 10.2.36
12.2.37	Definition	MC22007	Formerly Internet ET Standard No. 10.2.37
12.2.38	Definition	MC22007	Formerly Internet ET Standard No. 10.2.38
12.2.39	Definition	MC22007	Formerly Internet ET Standard No. 10.2.39
12.2.40	Definition	MC22007	Formerly QEDM Standard No. 4.2.19
12.2.41	Definition	MC22007	Formerly QEDM Standard No. 4.2.20
12.3.1	Standard	MC22007	Formerly Internet ET Standard No. 10.3.1
12.3.2	Standard	MC22007	Formerly Internet ET Standard No. 10.3.3
12.3.3	Standard	MC22007	Formerly Internet ET Standard No. 10.3.4
12.3.4	Standard	MC22007	Formerly Internet ET Standard No. 10.3.5
12.3.5	Standard	MC22007	Formerly Internet ET Standard No. 10.3.6
12.3.6	Standard	MC22007	Formerly Internet ET Standard No. 10.3.7
12.3.7	Standard	MC22007	Formerly Internet ET Standard No. 10.3.8
12.3.8	Standard	MC22007	Formerly Internet ET Standard No. 10.3.9
12.3.9	Standard	MC22007	Formerly Internet ET Standard No. 10.3.10
12.3.10	Standard	MC22007	Formerly Internet ET Standard No. 10.3.11
12.3.11	Standard	MC22007	Formerly Internet ET Standard No. 10.3.12
12.3.12	Standard	MC22007	Formerly Internet ET Standard No. 10.3.14
12.3.13	Standard	MC22007	Formerly Internet ET Standard No. 10.3.15
12.3.14	Standard	MC22007	Formerly Internet ET Standard No. 10.3.16
12.3.15	Standard	MC22007	Formerly Internet ET Standard No. 10.3.17
12.3.16	Standard	MC22007	Formerly Internet ET Standard No. 10.3.18
12.3.17	Standard	MC22007	Formerly Internet ET Standard No. 10.3.19
12.3.18	Standard	MC22007	Formerly Internet ET Standard No. 10.3.20
12.3.19	Standard	MC22007	Formerly Internet ET Standard No. 10.3.21
12.3.20	Standard	MC22007	Formerly Internet ET Standard No. 10.3.22
12.3.21	Standard	MC22007	Formerly Internet ET Standard No. 10.3.23
12.3.22	Standard	MC22007	Formerly Internet ET Standard No. 10.3.24
12.3.23	Standard	MC22007	Formerly Internet ET Standard No. 10.3.25
12.3.24	Standard	MC22007	Formerly Internet ET Standard No. 10.3.26
12.3.25	Standard	MC22007	Formerly Internet ET Standard No. 10.3.27
12.3.26	Standard	MC22007	Formerly Internet ET Standard No. 10.3.28
12.3.27	Standard	MC22007	Formerly Internet ET Standard No. 10.3.29
12.3.28	Standard	MC22007	Formerly QEDM Standard No. 4.3.84
12.3.29	Standard	MC22007 / R22003	Formerly QEDM Standard No. 4.3.60 Modified to incorporate Multi-Factor Authentication.
12.3.30	Standard	MC22007	Formerly QEDM Standard No. 4.3.62
12.3.31	Standard	MC22007	Formerly QEDM Standard No. 4.3.103
	Appendix A – Reference Guide	MC22007	Modified to incorporate additional security protocols from the QEDM Standards.
	Appendix B – Minimum Technical Characteristics and Guidelines for the Developer and User of the Customer Activities and Informational Postings Web Sites	MC2207	Formerly QEDM Appendix B

Standard	Description	Request No.	Action
	Appendix C – Minimum Technical Characteristics for EDM Communications	MC22007	Formerly QEDM Appendix C
	Appendix D – Frequently Asked Questions	MC22007	Formerly Internet ET Appendix B



3.0 November 14, 2014

The following table shows a summary of requests and interpretations resulting in modifications to the Version 3.0 WGQ Internet Electronic Transport Related Standards (equivalent to Retail Version 2.1).

Standard	Description	Request No.	Action
7.3.50	Interpretation	WGQ 2013 Annual Plan Item 5.b (attachment 1)	Delete interpretation
10.3.2	Standard	WGQ 2012 Annual Plan Item 14.a	Delete standard
Tab 6	Technical Implementation – Internet ET	R13005	Modify the language in the Acknowledgement Receipt: ‘gisb-acknowledgement-receipt’ and provide Example: Signed Content for Response with line numbers added for illustration only to provide clarification.

3.1 September 29, 2017

NO MODIFICATIONS WERE MADE TO THE VERSION 3.1 WGQ INTERNET ELECTRONIC TRANSPORT RELATED STANDARDS (EQUIVALENT TO RETAIL VERSION 3.2).

3.2 August 15, 2020

The following table shows a summary of requests and interpretations resulting in modifications to the Version 3.2 WGQ Internet Electronic Transport Related Standards (equivalent to Retail Version 3.3).

Standard	Description	Request No.	Action
	Table of Contents	MC18005	Minor correction to move the ‘Version Notes’ to the end of the manual and remove all references to ‘Tabs’ in the Table of Contents and Introduction in the NAESB WGQ Implementation Guides.
	Executive Summary	2019 WGQ Annual Plan Items 6.a and 6.b	Modify language in ‘Overview of Electronic Transport Life Cycle’ sub-section to specify transport layer security to address changes from Sandia National Laboratories surety assessment.
	Introduction	MC18005	Minor correction to remove all references to ‘Tabs’ in the Table of Contents, Introduction and Section Headings in the NAESB WGQ IET Implementation Guide.
	Business Process and Practices	2019 WGQ Annual Plan Items 6.a and 6.b	Modify language in ‘Data Privacy and Encryption’ sub-section to specify transport layer security to address changes from Sandia National Laboratories surety assessment.
	Business Process and Practices	2019 WGQ Annual Plan Items 6.a and 6.b	Modify language under the heading ‘A. Overview’, in ‘Authentication’ sub-section to specify transport layer security to address changes from Sandia National Laboratories surety assessment

Standard	Description	Request No.	Action
10.1.7	Principle	2019 WGQ Annual Plan Items 6.a and 6.b	Delete Principle
10.2.33	Definition	2019 WGQ Annual Plan Items 6.a and 6.b	Modify definition by changing 'SSL'. Secure Sockets Layer' to 'TLS'. Transport Layer Security to address changes from Sandia National Laboratories surety assessment.
10.2.34	Definition	2019 WGQ Annual Plan Items 6.a and 6.b	Modify definition to remove hyperlink reference
10.2.39	Definition	2019 WGQ Annual Plan Items 6.a and 6.b	Add new definition 'OpenPGP'. Open Pretty Good Privacy to address changes from Sandia National Laboratories surety assessment.
10.3.4	Standard	2019 WGQ Annual Plan Items 6.a and 6.b	Modify standard to address changes from Sandia National Laboratories surety assessment.
10.3.15	Standard	2019 WGQ Annual Plan Items 6.a and 6.b	Modify standard to address changes from Sandia National Laboratories surety assessment.
10.3.16	Standard	2019 WGQ Annual Plan Items 6.a and 6.b	Modify standard to specify transport layer security to address changes from Sandia National Laboratories surety assessment.
10.3.25	Standard	2019 WGQ Annual Plan Items 6.a and 6.b	Modify standard to specify transport layer security to address changes from Sandia National Laboratories surety assessment.
10.3.28	Standard	2019 WGQ Annual Plan Items 6.a and 6.b	Add new standard to address changes from Sandia National Laboratories surety assessment.
10.3.29	Standard	2019 WGQ Annual Plan Items 6.a and 6.b	Add new standard to address changes from Sandia National Laboratories surety assessment
	Technical Implementation - Internet ET	2019 WGQ Annual Plan Items 6.a and 6.b	Modify language in subsections 'Using an Interactive Browser for Internet ET' and 'Authentication' to specify transport layer security to address changes from Sandia National Laboratories surety assessment. Modify the 'EXAMPLE: AN X12 File Encrypted with PGP' –Begin PGP Message – by changing 'Version: PGP 6.5' to "Version: PGP 9.0" in all instances. Modify language in subsection 'Pre-validation before Decryption' to change 'PGP' to 'PGP/OpenPGP' Modify language in subsection 'Security' to change 'Open PGP' to 'OpenPGP' Modify language in subsections 'Understanding OpenPGP and PGP', 'Encryption / Digital Signature' and 'OpenPGP or PGP File Encryption' to address changes from Sandia National Laboratories surety assessment.

Standard	Description	Request No.	Action
	Testing Guidelines	2019 WGQ Annual Plan Items 6.a and 6.b	Modify language in subsection 'NAESB Internet TI Test Guidelines' to change 'PGP' to 'PGP/OpenPGP' to address changes from Sandia National Laboratories surety assessment.
	Appendix A	2019 WGQ Annual Plan Items 6.a and 6.b	Modify appendix language to specify transport layer security to address changes from Sandia National Laboratories surety assessment.

4.0 September 29, 2023

The following table shows a summary of requests and interpretations resulting in modifications to the Version 4.0 WGQ Internet Electronic Transport Related Standards

Standard	Description	Request No.	Action
	Executive Summary	MC22007	Minor correction to move from IET Related Standards, remove references to RMQ and Internet ET manual, and to add clarity to new purpose.
		2023 WGQ Annual Plan Item 1a and 1b	Modify to correct typographical errors
	Introduction	MC22007	Minor correction to move from IET Related Standards, remove references to RMQ and Internet ET manual, and to add clarity to new purpose
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
	Business Process and Practices	MC22007	Minor correction to move from IET Related Standards
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.1.1	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.1
12.1.2	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.2
12.1.3	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.3
12.1.4	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.4
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.1.5	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.5
12.1.6	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.6
12.1.7	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.8

Standard	Description	Request No.	Action
12.1.7		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.1.8	Principles	MC22007	Minor correction to move from IET Related Standard No. 10.1.10
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.1.9	Principles	MC22007	Minor correction to move from QEDM Related Standard No. 4.1.37
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.1	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.1
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.2	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.2
12.2.3	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.3
12.2.4	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.4
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.5	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.5
12.2.6	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.6
12.2.7	Definition	MC22007	Minor correction to move from IET Related Standards No. 10.2.7
12.2.8	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.8
12.2.9	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.9
12.2.10	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.10
12.2.11	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.11
12.2.12	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.12
12.2.13	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.13
12.2.14	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.14
12.2.15	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.15
12.2.16	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.16
12.2.17	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.17
12.2.18	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.18

Standard	Description	Request No.	Action
12.2.19	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.19
12.2.20	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.20
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Delete definition for 'Internet EDM'
12.2.21	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.21
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify by removing definition and mark as "[Deleted]"
12.2.22	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.22
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.23	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.23
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.24	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.24
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.25	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.25
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.26	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.26
12.2.27	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.27
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.28	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.28
12.2.29	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.29
12.2.30	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.30
12.2.31	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.31
12.2.32	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.32
12.2.33	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.33
12.2.34	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.34

Standard	Description	Request No.	Action
12.2.35	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.35
12.2.36	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.36
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.37	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.37
12.2.38	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.38
12.2.39	Definition	MC22007	Minor correction to move from IET Related Standard No. 10.2.39
12.2.40	Definition	MC22007	Minor correction to move from QEDM Related Standard 4.2.19
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.2.41	Definition	MC22007	Minor correction to move from QEDM Related Standard 4.2.20
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.1	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.1
12.3.2	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.3
12.3.3	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.4
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.4	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.5
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.5	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.6
12.3.6	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.7
12.3.7	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.8
12.3.8	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.9
12.3.9	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.10
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.10	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.11
12.3.11	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.12

Standard	Description	Request No.	Action
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.12	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.14
12.3.13	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.15
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.14	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.16
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
	Standard	MC23013	Modify to correct typographical errors
12.3.15	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.17
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.16	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.18
12.3.17	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.19
12.3.18	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.20
12.3.19	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.21
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.20	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.22
12.3.21	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.23
12.3.22	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.24
12.3.23	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.25
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.24	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.26
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.25	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.27
12.3.25		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors

Standard	Description	Request No.	Action
12.3.26	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.28
12.3.27	Standard	MC22007	Minor correction to move from IET Related Standard No. 10.3.29
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.28	Standard	MC22007	Minor correction to move from QEDM Related Standard No. 4.3.84
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.29	Standard	MC22007	Minor correction to move from QEDM Related Standard No. 4.3.60
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
12.3.30	Standard	MC22007	Minor correction to move from QEDM Related Standard 4.3.62
12.3.31	Standard	MC22007	Minor correction to move from QEDM Related Standard 4.3.103
	Appendix A	MC22007	Minor correction to move from IET Related Standards and incorporate additional security protocols from the QEDM Related Standards
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
	Appendix B	MC22007	Minor correction to move from the QEDM Related Standards
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
	Appendix C	MC22007	Minor correction to move from the QEDM Related Standards
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors
	Appendix D	MC22007	Minor correction to move from the IET Related Standards
		2023 WGQ Annual Plan Item 1a and 1b Attachment B	Modify to correct typographical errors